GRADIENT PROJECTION METHODS FOR CONTINUAL LEARNING IN HYPERNETWORKS

OVERCOMING CHUNK-INDUCED GRADIENT PATHOLOGIES VIA PROJECTION NORMALIZATION, AND THE INTRODUCTION OF IFOPNG

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The algorithmic framework of this thesis builds directly on the Fisher-Orthogonal Projected Natural Gradient method of Garg, Kolhe, Peng, and Gopalam (2026), whose derivations and design served as the primary technical foundation for `iFOPNG`. The hypernetwork architecture, weight-chunking scheme, and functional regularizer are due to Oswald, Henning, Grewe, and Sacramento (2020) and are adopted here.

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Abstract

This thesis addresses the fundamental incompatibility of applying gradient projection-based continual learning (CL) methods to hypernetwork architectures, establishing a novel framework for stable subspace optimization. Traditional projection methods collapse under hypernetwork architectures: the architectural necessity of weight-chunking forces the model to accumulate an exceptionally large gradient volume before a single projection step can execute, blowing up the condition number of the projection matrix and causing catastrophic inversion failures that standard regularisation cannot rescue.

The primary research objective is to eliminate these numerical singularities and systematically evaluate whether combining hard geometric projections on top of task-conditioned hypernetworks and their output-scoped functional regularizers yields a constructive synergy. To achieve this, we introduce *projection-scoped normalization*: gradient columns stored in the memory matrix G are rescaled to unit ℓ_2 norm, and the diagonal empirical Fisher vector is rescaled by its absolute maximum, together stabilizing the metric tensor within the local projection algebra. We also introduce `iFOPNG`, an inertial variant of Fisher-Orthogonal Projected Natural Gradient Descent (Garg et al., 2026) that replaces the current-task Fisher F_{new} with the combined curvature $F_c = F_{\text{new}} + F_{\text{old}}$, inspired by the parameter-inertia concept of Elastic Weight Consolidation. Methods are benchmarked on sequential Split and Permuted MNIST , Split-CIFAR10 and Split-CIFAR100. The empirical findings reveal that projection normalization resolves matrix inversion collapse in the tested configurations, and that `iFOPNG` consistently outperforms `FOPNG` across all benchmarks; in the standard-capacity hypernetwork setting only `iFOPNG` surpasses plain `SGD`, while most other projection variants fall below it. A standalone hypernetwork optimized with `Adam` and Oswald et

al.'s functional regularizer consistently surpasses all projection variants, indicating that momentum, which projection methods forgo by design, is the decisive factor in compressed generative weight spaces.

1 DATA SOURCE, ETHICS, CODE, AND TECHNOLOGY STATEMENT

Data Source. The MNIST (owned by LeCun et al.) and CIFAR (CIFAR-10 and CIFAR-100, owned by Krizhevsky et al.) datasets are public, anonymized datasets licensed for research. These datasets form the basis for the Split-MNIST, Permuted MNIST, Split-CIFAR10, and Split-CIFAR100 benchmarks evaluated in this work. No human or animal data was collected. Data was acquired via the `torchvision.datasets` API.

FIGURES. All figures and plots were created entirely by the author using `matplotlib` and Weights & Biases.

CODE. The baseline FOPNG framework was adapted from Garg et al. (2026); all other execution infrastructure was implemented by the author (`https://github.com/kelares/HyperFisher`). Libraries used: Python (v3.10), PyTorch (v2.x), Torchvision, Weights & Biases, NumPy, Matplotlib, and Tqdm.

TECHNOLOGY. The manuscript was typeset in LaTeX using BibTeX. During the preparation of this work, the author(s) used Gemini and Claude in order to optimize academic phrasing, outline structure, paraphrase transitions, and check grammar. After using these tools/services, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the thesis. AI Citations: Google. (2026). *Gemini* [Large language model]. https://gemini.google.com; Anthropic. (2026). *Claude* [Large language model]. https://claude.ai.

2 INTRODUCTION

The capacity for continuous, sequential adaptation without the eradication of historical knowledge remains an architectural cornerstone of biological intelligence, yet it presents a fundamental barrier for artificial neural networks. In deep learning paradigms, sequential training across non-stationary data distributions triggers a systemic pathology known as *catastrophic forgetting* (McCloskey & Cohen, 1989). When a network optimizes its parameters for an incoming task, gradient updates blindly overwrite the weight configurations essential for retaining prior task spaces, shifting the internal representation manifolds away from historical objective minima.

Beyond its theoretical interest, this limitation carries direct practical cost. Deployed systems must either be retrained from scratch whenever the data distribution shifts — at substantial computational and energetic expense — or retain growing archives of historical data for replay, an option foreclosed in privacy-regulated and on-device settings where past data cannot be stored. Continual learning methods that preserve prior competence without replay therefore address both the economic burden of repeated retraining and the deployment of adaptive models under data-retention constraints. The societal stakes are concrete: personal health monitoring devices must adapt to a patient's evolving physiology without transmitting historical biometric data to external servers; edge-deployed vision systems in vehicles or industrial equipment must absorb distribution shifts in-situ without the energy and latency cost of cloud retraining; and large-scale language systems must incorporate new knowledge without the carbon footprint of full retraining cycles. The normalization and inertial-metric contributions of this thesis lower a foundational barrier to deploying such systems under strict data-retention and compute constraints.

Modern continual learning (CL) literature is conventionally structured into three core algorithmic families: replay-based strategies that leverage episodic data buffers, regularization-based frameworks that penalize parameter deviations, and architectural or parameter-isolation methods (De Lange et al., 2022). Within the meta-learning domain, hypernetworks (Ha, Dai, & Le, 2017) have emerged as an elegant paradigm. Hypernetworks circumvent parameter-space conflicts by utilizing a centralized, lightweight meta-model to generate the target network's complete parameter weights conditioned on task embeddings, as formalized by Oswald et al. (2020). Concurrently, within the field of geometric optimization, gradient projection methods, exemplified by the recently proposed Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026), explicitly constrain subsequent updates to lie orthogonal to the subspace of parameter directions critical for previous tasks, with the geometry of orthogonality ranging from Euclidean (OGD) to Fisher-Riemannian (FOPNG).

Despite the mathematical elegance of both hypernetworks and gradient projection methods, their structural intersection uncovers a critical and previously unexamined optimization pathology. Generating a target network's full weight vector in one forward pass is computationally prohibitive, so chunked hypernetworks partition generation into K sequential sub-passes from a shared generator. Because every sub-pass executes before any gradient update, the gradients and Fisher estimates accumulated across chunks inflate with K, rendering the projection matrix numerically uninvertible. The exact algebraic mechanism is given in Section 4.5.1.

Research Objectives and Algorithmic Framework

The primary research objective of this thesis is to systematically resolve this linear-algebraic collapse, map the initialization mechanics of meta-projection pipelines, and evaluate whether the integration of hard geometric constraints on top of soft functional regularizers yields an optimized sequential learning system.

To achieve this, we introduce *projection-scoped normalization* (column-wise ℓ_2 orthonormalization of G combined with absolute-maximum scaling of $\tilde{F}$) and `iFOPNG`, an inertial FOPNG variant using the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$. These contributions enable stable gradient projection in chunked hypernetworks, with direct applicability to on-device sequential learning where retaining adaptive momentum is critical.

Research Questions

To systematically navigate these boundaries, this thesis addresses the following primary research question:

> *What numerical pathologies arise when gradient projection methods are applied to chunked hypernetwork architectures, and do modifications to the Fisher metric construction—specifically an inertial combined metric and a monotonic accumulation operator—yield principled improvements over standard projection baselines in continual learning?*

This is decomposed into four sub-research questions (Sub-RQs):

Sub-RQ1 **(Synergy vs. Conflict):** Does the joint integration of hard gradient projection with the hypernetwork's soft functional regularizer yield a constructive synergy, or does a standalone regularized hypernetwork offer a more efficient optimization frontier?

Sub-RQ2 **(Gradient Pathology):** How does the architectural necessity of spawning K sequential weight-chunks inflate the gradient and Fisher magnitudes accumulated before a single projection step, and does rescaling both components independently within the projection algebra suffice to prevent matrix inversion collapse?

Sub-RQ3 **(Parameter Inertia):** Does replacing `FOPNG`'s task-specific metric F_{new} with the combined inertial metric $F_c = F_{\text{new}} + F_{\text{old}}$ yield consistent retention improvements across benchmarks?

Sub-RQ4 **(Fisher Accumulation):** Does accumulating F_{old} via a non-decaying element-wise maximum operator maintain long-horizon retention more effectively than an exponential moving average, and does the monotonic growth of the maximum operator impose a measurable plasticity cost over extended task sequences?

Summary of Empirical Findings

Projection-scoped normalization fully resolves the matrix inversion pathology, recovering stable projection performance across all seeds. The inertial combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ consistently improves retention over `FOPNG`, with the advantage scaling monotonically with inter-task Fisher overlap. The MAX accumulation operator outperforms EMA on long task sequences and requires no additional hyperparameters. In the standard-capacity hypernetwork setting, `iFOPNG` surpasses plain `SGD` but falls short of `Adam`; under extreme compression the gap to `Adam` widens to 15–30 pp and most projection methods drop below `SGD` entirely.

3 RELATED WORK

3.1 *Continual Learning and the Stability-Plasticity Dilemma*

Modern machine learning systems are predominantly trained on fixed datasets drawn from a single stationary distribution — a condition that rarely holds in deployment, where agents must adapt to non-stationary data streams while retaining prior knowledge. The fundamental tension between these two objectives is formalized as the *stability-plasticity dilemma* (Grossberg, 1980): an agent must remain plastic enough to absorb novel distributions yet stable enough to resist the erasure of established representations.

The continual learning literature addresses this across three canonical scenarios (De Lange et al., 2022): *task-incremental learning*, where the task identity is available at both training and test time; *class-incremental learning*, where task identity is withheld at inference; and *domain-incremental learning*, where the output space is shared but the input distribution shifts continuously. This thesis focuses on the task-incremental setting, the scenario in which projection-based methods and hypernetwork architectures have been most thoroughly evaluated and in which structural comparisons between them are most tractable.

3.2 *Evaluation Protocols: Multi-Head and Single-Head Designs*

Task-incremental benchmarks are evaluated under one of two output-layer protocols. The *multi-head* protocol gives each task a dedicated output head, selected by task identity at test time, which removes output-level interference between tasks; the *single-head* protocol shares one output layer, forcing the backbone alone to resolve that interference. Farquhar and Gal (2018) showed that multi-head evaluation systematically inflates apparent method performance by eliminating a major source of forgetting, and argued that the single-head protocol is the more rigorous test. Split-MNIST is therefore evaluated under both: the multi-head variant reproduces the conditions of Garg et al. (2026) for direct numerical comparison, and the single-head variant follows Farquhar and Gal (2018) as the harder, more discriminating evaluation. The exact head configuration and the protocol assigned to each benchmark are given in Section 4.6.1.

3.3 *Fisher Information Matrix*

Several methods surveyed in this thesis quantify parameter importance through the *Fisher Information Matrix* (FIM). For a parametric model $p_\theta(y \mid x)$ with parameters $\theta \in \mathbb{R}^P$, the FIM is the $P \times P$ matrix

$$F(\theta) = \mathbb{E}_{x\sim q(x)}\, \mathbb{E}_{y\sim p_\theta(\cdot|x)} \Big[\nabla_\theta \log p_\theta(y \mid x)\, \nabla_\theta \log p_\theta(y \mid x)^\top\Big], \tag{1}$$

where $q(x)$ is the data-generating distribution (Amari, 1998). $F(\theta)$ defines a Riemannian metric on the parameter manifold: the Fisher inner product $\langle u, v\rangle_F = u^\top F(\theta)\, v$ measures distances in terms of the change in model predictions, not in terms of raw Euclidean parameter displacement.

The full FIM is intractable at the scale of neural networks. All methods evaluated in this thesis use the *diagonal empirical approximation*, which replaces the outer-product expectation with an empirical average over N observed data points:

$$\hat{F}_i \;\approx\; \frac{1}{N} \sum_{n=1}^{N} \left(\frac{\partial \mathcal{L}^{(n)}}{\partial \theta_i}\right)^2, \tag{2}$$

yielding a non-negative scalar $\hat{F}_i$ for each parameter θ_i. The resulting vector $\hat{F} \in \mathbb{R}^P_{\geq 0}$ is treated as a diagonal matrix in all subsequent quadratic forms. This approximation reduces storage from $\mathcal{O}(P^2)$ to $\mathcal{O}(P)$ and is the variant used by EWC (Kirkpatrick et al., 2017) and all Fisher-based projection methods evaluated in (Garg et al., 2026).

3.4 *Regularization-Based Approaches*

The dominant paradigm for addressing catastrophic forgetting is *parameter regularization*, of which Elastic Weight Consolidation (EWC) (Kirkpatrick et al., 2017) is the canonical representative. EWC uses F_i (Equation 2) as an importance weight for each parameter, and applies a soft quadratic penalty that resists changes to parameters deemed important to prior tasks:

$$\mathcal{L}_{\text{EWC}}(\theta) = \mathcal{L}_t(\theta) + \frac{\lambda}{2} \sum_i F_i \big(\theta_i - \theta^*_{t-1,i}\big)^2, \tag{3}$$

where $\theta \in \mathbb{R}^p$ are the model parameters, θ^*_{t-1} is their snapshot saved after task $t-1$, and $\lambda > 0$ governs overall regularization strength. EWC is computationally efficient and broadly applicable, and serves as the standard reference point against which new continual learning methods are measured. Its principal limitation is that soft penalties permit bounded parameter drift: cumulative updates across a long task sequence gradually erode the protection of earlier task spaces, a failure mode that becomes increasingly pronounced as T grows (De Lange et al., 2022).

3.5 *Gradient Projection Methods*

To enforce strict, non-negotiable protection for prior knowledge rather than relying on penalties that can be overcome by sufficiently large gradients, a second paradigm uses explicit *gradient projection*. Orthogonal Gradient Descent (OGD) (Farajtabar, Azizan, Mott, & Li, 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of historical task gradient directions and projects incoming gradients onto the orthogonal complement of $\text{Col}(G)$ before each parameter update:

$$\tilde{g} = g - G(G^\top G)^{-1} G^\top g. \tag{4}$$

This provides a hard guarantee: the projected update cannot increase the loss on any previously seen task. Generalization bounds established within the Neural Tangent Kernel regime confirm that OGD provides exact protection in linear and mildly nonlinear settings (Bennani, Doan, & Sugiyama, 2020).

However, standard Euclidean projection treats the parameter space as flat, assigning equal importance to all directions regardless of the curvature of the loss surface.

Fisher-Orthogonal Projected Natural Gradient Descent (FOPNG) (Garg et al., 2026) addresses this by performing the projection in the Riemannian geometry defined by $\hat{F}$, the diagonal empirical Fisher of the current task, so that the orthogonality constraint respects the local curvature of the loss surface. The related Fisher Natural Gradient (FNG) (Garg et al., 2026) applies the natural gradient without any projection, and Orthogonal Natural Gradient (ONG) (Yadav, Mendoza, & Korrapati, 2025) combines Fisher preconditioning with Euclidean projection, sitting structurally between OGD and FOPNG. Exact algorithmic instantiations are given in Section 4.3.

3.6 *Hypernetworks for Continual Learning*

An orthogonal architectural response to catastrophic forgetting replaces the multi-head output design entirely with a *hypernetwork* (Ha et al., 2017): a meta-model $\mathcal{H}(\cdot;\phi)$ trained to synthesize the complete weight vector $\theta_t = \mathcal{H}(e_t;\phi)$ of a target network, conditioned on a task embedding e_t. Each task receives a unique weight configuration generated on demand, rather than a dedicated output head, achieving implicit task separation through the generative bottleneck.

Generating the full θ_t in a single forward pass is computationally intractable for large target networks, so Oswald et al. (2020) introduced *weight chunking*: the generator iterates over C chunk embeddings $c_1, \dots, c_C$, producing a fixed-size block of target parameters per pass and concatenating them into the full weight vector θ_t. To prevent the hypernetwork from forgetting how to generate the weights of earlier tasks, a soft *functional regularizer* anchors the generator in its own output space. Let $\theta_s = \mathcal{H}(e_s;\phi)$ be the target-network weight vector the generator currently produces for task s under the single, evolving parameter set ϕ. On completion of task s, the generated vector is cached as a fixed reference θ_s^{ref}, detached from the computation graph. While task t is trained, only ϕ is updated, and every reference $\{\theta_s^{\text{ref}}\}_{s<t}$ stays fixed at the value it held when its task finished, under the penalty

$$\mathcal{L}_{\text{reg}} = \frac{\beta}{t-1} \sum_{s=1}^{t-1} \left\| \mathcal{H}(e_s;\phi) - \theta_s^{\text{ref}} \right\|_2^2, \tag{5}$$

with regularization strength $\beta > 0$. The penalty therefore measures, in the space of generated weights, how far the current generator has drifted from the exact target parameters it once produced for each past task. This can be interpreted as EWC applied at the meta level: rather than anchoring individual target parameters, it anchors the generated weight configurations for each previously seen task, penalising drift in the function space where task-relevant quantities actually live.

3.7 *Identified Gaps and Motivating Pathologies*

Whether hard geometric projection constraints are compatible with the soft functional regularization already employed by hypernetworks, or whether layering both simultaneously over-constrains the generator's plasticity, has not been investigated (Sub-RQ1).

The architectural necessity of weight-chunking introduces a gradient accumulation pathology — inflated condition numbers in the projection kernel — that is unique to the hypernetwork and projection methods combination and has no prior treatment in the literature (Sub-RQ2).

Existing Fisher-based projection methods define the projection geometry from a single-task Fisher metric, which does not weight parameters by their importance to past tasks; whether replacing it with a combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ that anchors historically important coordinates, giving them inertia, yields consistent retention improvements across benchmarks has not been examined (Sub-RQ3).

The historical Fisher estimate underlying Fisher-based projection depends on a choice of accumulation operator; whether a monotonic maximum or a decaying exponential moving average better preserves long-horizon retention has not been directly compared (Sub-RQ4).

4 METHODS

4.1 *Continual Learning Protocol*

4.1.1 *Task Setup and Evaluation Metrics*

A continual learning agent receives a sequence of T tasks $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_T$ presented in order. At task t, only the data from $\mathcal{T}_t$ is available; prior task data is not retained. After each task is trained, the model is evaluated on all tasks seen so far. We report two aggregate metrics: *average accuracy*

$$A_t = \frac{1}{t} \sum_{i=1}^{t} a_{t,i}, \tag{6}$$

and *backward transfer*

$$\mathrm{BWT} = \frac{1}{T-1} \sum_{t=2}^{T} (a_{T,t} - a_{t,t}), \tag{7}$$

where $a_{i,j}$ denotes accuracy on task j immediately after training task i. Negative BWT quantifies catastrophic forgetting; a value of zero indicates perfect retention.

Reporting both metrics jointly is necessary because each can mask failure modes invisible to the other: a method that forecloses new learning to preserve old tasks achieves near-zero BWT alongside collapsed accuracy, while a method that overfits each new task sustains competitive accuracy at the cost of large negative BWT on earlier ones.

4.2 *Fisher Information Matrix and Regularization-Based Continual Learning*

All Fisher-based methods evaluated in this thesis use the diagonal empirical approximation F defined in Equation (2). The F_{new} and F_{old} used in projection methods denote the FIMs computed from the current task's data and accumulated from all prior tasks, respectively; both are instances of Equation (2).

4.2.1 *Elastic Weight Consolidation*

EWC (Section 3.4, Equation 3) is included as a regularization baseline that operates purely through parameter-space penalties and maintains no gradient subspace. We use $\lambda = 10^{-3}$ for all hypernetwork configurations and reproduce the per-benchmark values from Garg et al. (2026) Table 1 for standalone evaluations. Fisher estimates are computed over the full training set for MNIST-based benchmarks and over N=1024 samples for CIFAR benchmarks.

4.3 *Gradient Projection Methods*

This subsection presents the full suite of gradient-based methods evaluated in this thesis. `SGD` and `FNG` establish unconstrained lower bounds with no forgetting-protection mechanism; `OGD` through `iFOPNG` form the projection family that is the primary focus of this work.

4.3.1 *Stochastic Gradient Descent*

`SGD` applies vanilla gradient descent with no forgetting protection:

$$\Delta\theta_{\text{SGD}} = -\eta \, \nabla_\theta \mathcal{L}_t. \tag{8}$$

`Adam` (Kingma & Ba, 2015) is included as an adaptive-rate variant that maintains per-parameter momentum; both serve as unconstrained baselines.

4.3.2 *Orthogonal Gradient Descent*

OGD (Farajtabar et al., 2020) maintains a memory matrix $G = [g_1 \mid \cdots \mid g_K] \in \mathbb{R}^{p \times K}$ of past task gradient directions and updates parameters via the current gradient projected onto the orthogonal complement of $\mathrm{Col}(G)$:

$$\tilde{g} = g - G(G^\top G)^{-1}G^\top g, \qquad \Delta\theta_{\mathrm{OGD}} = -\eta\,\tilde{g}. \tag{9}$$

GRADIENT MEMORY MANAGEMENT. In all projection methods, G is maintained as a fixed-budget basis of at most M columns; insertion, orthonormalisation, and overflow recompression are detailed in Section 4.5.1.

4.3.3 *Natural Gradient and Fisher Natural Gradient*

Natural gradient descent (Amari, 1998) replaces the Euclidean gradient step with one measured in the Riemannian geometry of the Fisher metric, so that equal step sizes correspond to equal changes in model predictions rather than equal changes in parameter values:

$$\Delta\theta_{\mathrm{nat}} = -\eta\, F_{\mathrm{new}}^{-1}\nabla_\theta \mathcal{L}. \tag{10}$$

`FNG` (Garg et al., 2026) applies a trust-region constraint to this step, normalising by the Fisher norm of the gradient:

$$v_{\mathrm{FNG}}^{*} = -\eta\,\frac{F_{\mathrm{new}}^{-1}g}{\sqrt{g^\top F_{\mathrm{new}}^{-1}g}}. \tag{11}$$

FNG maintains no gradient subspace and provides no forgetting protection; it is included to isolate the contribution of Fisher preconditioning independently of projection.

4.3.4 *Orthogonal Natural Gradient*

ONG (Yadav et al., 2025) extends OGD by replacing the raw gradient with the natural gradient before projection. The natural gradient $F_{\mathrm{new}}^{-1}g$ is first computed, then projected onto the Euclidean orthogonal complement of $\mathrm{Col}(G)$:

$$\Delta\theta_{\mathrm{ONG}} = -\eta\left(F_{\mathrm{new}}^{-1}g - G(G^\top G)^{-1}G^\top F_{\mathrm{new}}^{-1}g\right). \tag{12}$$

The step direction is Fisher-preconditioned, but the orthogonality constraint is still measured in Euclidean space: the kernel being inverted is $A = G^\top G + \lambda I$, identical to OGD. Following the diagonal Fisher approximation used throughout this thesis, we substitute the EKFAC inverse of Yadav et al. (2025) with the diagonal empirical Fisher; results are not directly comparable to the original implementation.

4.3.5 *Fisher-Orthogonal Projected Natural Gradient Descent*

`FOPNG` (Garg et al., 2026) moves both the step and the projection into the Riemannian geometry of the Fisher metric. Where ONG projects the natural gradient in Euclidean space, FOPNG measures orthogonality in Fisher space: a direction is orthogonal to the stored basis if it produces no change in model predictions along those directions, not merely no change in parameter values. The constrained trust-region update is:

$$v^*_{\text{FOPNG}} = -\eta \, \frac{F_{\text{new}}^{-1} P g}{\sqrt{(Pg)^\top F_{\text{new}}^{-1} P g}}, \qquad P = I - F_{\text{old}}\, G \Big(G^\top F_{\text{old}}\, F_{\text{new}}^{-1} F_{\text{old}}\, G + \lambda I \Big)^{-1} G^\top F_{\text{old}}, \tag{13}$$

where F_{new} is the current-task empirical Fisher, F_{old} is the accumulated historical Fisher under which the stored directions were collected, and λI is a damping term for numerical stability. This is the pre-Fisher form of Garg et al. (2026), in which the stored basis carries its collection-time Fisher weighting.

4.3.6 *Inertial FOPNG*

In FOPNG the Riemannian metric governing both the trust-region constraint and the projection objective is F_{new} alone. This is natural on the first task but creates an asymmetry as history accumulates: the subspace $\text{Col}(G)$ encodes directions important to prior tasks, yet the metric assigns zero curvature to parameters that were important previously but are uninformative on the current task, allowing the update to move freely through historically sensitive directions.

`iFOPNG` resolves this asymmetry by replacing F_{new} with the *combined metric*

$$F_c \;:=\; F_{\text{new}} + F_{\text{old}}, \tag{14}$$

where F_{old} is maintained as the element-wise maximum of all per-task Fisher diagonals accumulated so far. Substituting F_c for F_{new} throughout Equation (13) yields:

$$v^* = -\eta \, \frac{F_c^{-1} P g}{\sqrt{(Pg)^\top F_c^{-1} P g}}, \qquad P = I - F_{\text{old}}\, G\, A^{-1}\, G^\top F_{\text{old}}, \tag{15}$$

where $A = G^\top F_{\text{old}}\, F_c^{-1}\, F_{\text{old}}\, G + \lambda I$. This update is derived as the closed-form solution of the trust-region projection problem in Theorem 1 (Appendix 8).

The combined metric assigns each coordinate i an effective curvature $(F_{\text{new}})_i + (F_{\text{old}})_i$, suppressing the step size in directions important to any task in the sequence. We call this *parameter inertia*: a coordinate that accumulated high Fisher importance is not pulled back toward a stored reference point — it is made harder to displace. This contrasts with EWC (Kirkpatrick et al., 2017), which achieves retention through an explicit quadratic penalty anchored at θ^*; `iFOPNG` achieves it geometrically, through the projection metric alone.

4.4 *Hypernetwork Architecture*

4.4.1 *Chunked Hypernetworks*

Generating θ_t in a single forward pass would require an output layer of dimension $|\theta_t|$, which is computationally prohibitive for any non-trivial target architecture. *Weight chunking* (Oswald et al., 2020) resolves this by iterating over a secondary chunk embedding c_k, $k = 1, \ldots, K$, and producing a block of C_s target parameters per pass; the full weight vector is assembled by concatenation. Chunking makes the hypernetwork parameter-efficient but introduces the chunk-induced gradient accumulation pathology addressed in Section 4.5.1.

4.4.2 *Functional Regularizer*

The functional regularizer of Oswald et al. (2020) is active for all hypernetwork configurations, with regularization strength β treated as a tunable hyperparameter (see Appendix 8).

4.4.3 *Target Network and Dropout*

Standalone configurations reproduce the target network architecture of Garg et al. (2026) without modification, including two `Dropout`$(p{=}0.5)$ layers in the `SimpleCIFARCNN` classifier head. Dropout is removed from the target network in all hypernetwork configurations. In chunked generation, the hypernetwork executes K sequential forward passes through the same target architecture before any gradient update; because all passes share the parent module's training state, stochastic dropout masks are resampled independently on every chunk pass, injecting uncontrolled noise into the gradient signal accumulated across chunks. Removing dropout eliminates this source of variance without modifying any other aspect of the target architecture

4.5 *Novel Algorithmic Contributions*

4.5.1 *Projection-Scoped Normalization*

When a hypernetwork spawns K sequential weight-chunks before a single projection step executes, the accumulated gradient vector and the diagonal Fisher entries grow in magnitude proportionally to K, inflating the condition number of the projection kernel A and causing the matrix inversion in Equation (13) to fail catastrophically for any practically sized target network.

We resolve this with two independent rescaling operations that address distinct sources of ill-conditioning:

1. **Gradient basis orthonormalization.** Each batch of K gradient vectors collected after task t is replaced by the orthonormal factor Q from its thin QR decomposition before insertion into G:

$$[g_{t,1} \mid \cdots \mid g_{t,K}] \xrightarrow{\text{QR}} Q, \qquad Q^\top Q = I. \tag{16}$$

   QR is chosen over SVD at insertion time because the operation requires only orthonormalization of the incoming batch — the number of columns is preserved, not reduced, so the ordered factorization that SVD provides is unnecessary overhead. SVD is reserved for the overflow handler: when the column count of G exceeds the budget M, the full basis is recompressed as $G \leftarrow U_{:,1:M}$, retaining the M left singular vectors of greatest variance and discarding the rest. This operation applies *at storage time* and affects only the basis G; the gradient g entering the projection update itself is the unmodified parameter gradient from the model.

2. **Metric-tensor scaling.** Within each projection computation, all Fisher quantities are jointly divided by a single scale factor derived from the maximum entry of the method's ambient metric F^*:

$$s = \max\big(F^*,\, 1\big), \qquad \tilde{F} = F \,/\, s \;\text{ for } F \in \big\{F_{\text{old}}, F_{\text{new}}, F_c\big\}, \qquad \tilde{\lambda} = \lambda \,/\, s, \tag{17}$$

   where $F^* = F_{\text{new}}$ for `FNG`, `ONG`, and `FOPNG`, and $F^* = F_c$ for `iFOPNG`. The clamp to 1 prevents division below unity when the Fisher is near-zero. The damping coefficient λ is rescaled by the same factor, so its ratio to the Fisher magnitude is held fixed as the absolute Fisher scale drifts across tasks. This is necessary because a single additive λ cannot regularise the kernel by itself once chunk accumulation has inflated its spectrum: with the smallest and largest eigenvalues of A separated by as much as ten orders of magnitude (Section 5.2), any λ large enough to lift the smallest eigenvalues away from zero simultaneously swamps the informative mid-spectrum directions, while any λ small enough to leave those directions intact is negligible against the largest eigenvalues and does nothing for the conditioning. Because the condition number is invariant to a positive scalar multiple of A, the rescaling does not reduce $\kappa(A)$ directly; its purpose is to return the absolute magnitudes of A, its damped inverse, and the trust-region denominator to a range in which finite-precision arithmetic retains its significant digits and the fixed λ acts as a meaningful regulariser rather than as numerical noise.

Together, these operations ensure that the method-specific projection kernel A remains well-conditioned regardless of the number of chunks spawned per task. The kernel differs structurally across methods:

$$A = \begin{cases} G^\top G + \lambda I & \texttt{OGD, ONG} \\ G^\top F_{\text{old}}\, F_{\text{new}}^{-1}\, F_{\text{old}}\, G + \lambda I & \texttt{FOPNG} \\ G^\top F_{\text{old}}\, F_c^{-1}\, F_{\text{old}}\, G + \lambda I & \texttt{iFOPNG} \end{cases} \tag{18}$$

Gradient orthonormalization eliminates column-scale inflation in G, which directly reduces the spectral range of A for all three variants. Metric-tensor normalization bounds $\tilde{F}$ to $[0, 1]$, preventing the Fisher-weighted variants from producing numerically singular A. The `OGD/ONG` kernel is unaffected by metric-tensor scaling by construction. Subspace membership is invariant to positive scalar rescaling of basis vectors, so neither operation alters the geometry of the projection.

4.6 *Experimental Design*

The four sub-research questions demand qualitatively different experimental designs. Sub-RQ1 and Sub-RQ3 require head-to-head benchmark comparisons — projection methods against an unconstrained baseline, and `iFOPNG` against `FOPNG` respectively. Sub-RQ2 requires a controlled ablation over normalization conditions. Sub-RQ4 requires a long-horizon comparison of two Fisher accumulation operators on the same inertial combined-metric preconditioner.

4.6.1 *Benchmarks*

Experiments are conducted on four standard continual learning benchmarks, following the selection of Garg et al. (2026) to enable direct numerical comparison.

Permuted-MNIST (Kirkpatrick et al., 2017) applies a distinct fixed random pixel permutation per task to a shared digit-classification head, evaluated as a 5-task stream in the standalone suite (Sub-RQ3) and as a 20-task long-horizon stream for the accumulation-operator comparison (Sub-RQ4).

Split-MNIST splits the 10-digit dataset into 5 binary tasks of 2 classes each; the multi-head protocol follows Zenke, Poole, and Ganguli (2017) and the single-head variant follows Ven and Tolias (2019), constituting the harder evaluation.

Split-CIFAR10 (Zenke et al., 2017) splits CIFAR-10 into 5 two-class tasks under a multi-head protocol, providing genuine visual inter-task interference.

Split-CIFAR100 (Zenke et al., 2017) extends this to 10 tasks of 10 classes each, testing scalability under a multi-head protocol. Hypernetwork configurations are evaluated on Split-MNIST, Split-CIFAR10, and Split-CIFAR100 with the architectural modifications described in Section 4.4.

4.6.2 *Hyperparameter Provenance*

STANDALONE TARGET-NETWORK EXPERIMENTS. All hyperparameter values for experiments without a hypernetwork — learning rates, regularization coefficients, gradient budgets, Fisher sample counts, batch sizes, and epoch counts — are taken without modification from Garg et al. (2026) Table 1. `iFOPNG` uses the same values as `FOPNG`, since the inertial combined Fisher introduces no hyperparameters beyond those FOPNG already defines. `ONG` (Yadav et al., 2025) is treated as a composite method: it applies `OGD`'s Euclidean projection operator to the natural gradient direction and introduces no hyperparameters beyond those already specified for `OGD` and `FNG`; its values therefore follow `OGD`'s Table 1 entries.

DEVIATION FOR FIRST-TASK INITIALIZATION. We deviated from the original protocol in one specific instance: first-task initialization for MNIST-based benchmarks. While Garg et al. (2026) state they used the method's own learning rate for first-task SGD training (e.g., $\eta = 10^{-5}$ for FOPNG), our empirical replication of this protocol yielded an initial accuracy of only $\approx 21\%$. As demonstrated by the learning rate convergence sweep detailed in Appendix D (Figure 11), this published rate falls well below the empirical convergence threshold, effectively failing to establish a functional baseline. To ensure a clean initial projection subspace, we standardized the first-task SGD learning rate to 10^{-3} for MNIST-based benchmarks and 10^{-2} for CIFAR-based benchmarks. Beyond this necessary empirical correction, no tuning was performed on the published values.

HYPERNETWORK EXPERIMENTS. Settings not covered by Garg et al. (2026) follow the canonical hypernetwork framework of Oswald et al. (2020); exact values are detailed in Appendix 8. Because computing a full Fisher estimate requires a complete K-chunk generation pass per training sample, estimation is capped at 1,024 samples to maintain computational tractability — a standard approximation consistent with prior CIFAR-10 evaluations (Kirkpatrick et al., 2017). The gradient budget remains $K = 80$ per task, with memory compression handled via QR-and-SVD (Section 4.3). First-task training uses `AdamW` to seed the memory matrix G and Fisher estimate F from a clean geometric basis.

Because no prior work evaluates these projection methods inside a chunked hypernetwork, the hypernetwork results are not benchmarked against any external baseline and the absolute accuracies are not directly comparable to the standalone numbers or to those of Garg et al. (2026). They are instead governed by three internally controlled choices: a uniform learning rate of 10^{-3} across all methods, the functional regularizer of Oswald et al. (2020) active throughout, and a generator capacity selected from the architecture ablation in Appendix 8. The comparisons are therefore valid only within this fixed configuration, where the projection geometry is the sole independent variable.

TRAINING DURATION AND SCHEDULING. Per-task training terminates at the first of: loss falling below 0.1; ten consecutive non-improving epochs; or the benchmark epoch cap (Appendix 8). A learning rate scheduler halves the magnitude after every three non-improving epochs; the best-loss parameter state is restored at termination.

RANDOM SEEDS. All experiments draw from a fixed pool of random seeds. We allocate seeds as follows:

- **Standard configurations (**$n = 5$**):** $\{42, 111, 811, 1234, 2137\}$.
- **Split-CIFAR10 hypernetwork (**$n = 6$**):** Adds seed 314 to the standard set.
- **Split-MNIST hypernetwork (**$n = 5$**):** Replaces seed 2137 with 314, yielding $\{42, 111, 314, 811, 1234\}$. This substitution is applied uniformly across all methods to ensure a fair comparison, as seed 2137 caused a degenerate initialization exclusively for projection-based methods while leaving unconstrained baselines unaffected.

FISHER ACCUMULATION DEFAULT. The EMA decay parameter α appears exclusively in the Sub-RQ4 operator comparison ($\alpha = 0.5$); every other experiment accumulates F_{old} with the element-wise MAX operator, which introduces no additional hyperparameter.

STATISTICAL ANALYSIS. Pairwise comparisons (Sub-RQ3: `iFOPNG` vs. `FOPNG`; Sub-RQ4: MAX vs. EMA) use a two-tailed paired t-test over per-seed final average accuracy, with significance at $\alpha = .05$. Effect size is reported as Cohen's $d_z = \bar{D}/s_D$, the mean per-seed difference divided by its sample standard deviation, so large values reflect seed-to-seed consistency rather than raw magnitude. The full eight-method comparison is reported descriptively rather than tested.

4.6.3 *Sub-RQ1 — Constructive Synergy or Architectural Conflict*

EXPERIMENTAL DESIGN. Sub-RQ1 investigates whether layering a hard geometric projection on top of a hypernetwork's soft functional regularizer yields a net benefit, or if the loss of adaptive momentum makes a standard `Adam`-regularized hypernetwork superior. To separate architectural conflicts from benchmark-specific artifacts, we evaluate across three datasets at varying levels of compression, comparing an unconstrained standalone network (Panel a) against a hypernetwork (Panel b).

BENCHMARK SELECTION. We utilize a three-benchmark progression. **Benchmark 1 (Split-MNIST)** utilizes a deliberately "suffocated" hypernetwork (generator hidden dimension $d_h = 8$) to expose differential effects without hitting the near-perfect accuracy ceilings observed at higher capacities (see Appendix 8, Figure 8). To ensure convergence under these constraints, each task is trained for 15 epochs with a batch size of 64. **Benchmark 2 (Split-CIFAR10)** serves as the primary test, offering genuine inter-task interference and utilizing a standard-capacity hypernetwork trained for 50 epochs. **Benchmark 3 (Split-CIFAR100)** extends this sequence to 10 tasks to test scalability. Comparing performance within and across these panels reveals whether projection methods universally dominate, conditionally succeed based on compression, or systematically fail due to momentum stripping.

4.6.4 *Sub-RQ2 — Projection-Scoped Normalization*

WHAT WE WANT TO DIFFERENTIATE. Sub-RQ2 asks whether the chunk-induced gradient accumulation pathology is real, characterisable, and resolvable. A single accuracy number cannot answer this: we need direct numerical evidence that the pathology occurs without normalization, that normalization resolves it, and that the resolution does not compromise projection quality.

EXPERIMENTAL DESIGN. `iFOPNG` is evaluated under three normalization conditions on the Split-CIFAR10 hypernetwork (Benchmark 2), with all other hyperparameters held constant. Split-CIFAR10 is chosen over the CIFAR-100 and MNIST hypernetwork configurations specifically because its chunk size of 6 000 yields $C = 197$ chunks per target network — large enough to produce a measurable pathology, yet small enough that the condition number of A can be computed and logged at every projection step without becoming numerically intractable.

CONDITION 1 — NO NORMALIZATION (NEGATIVE CONTROL). Incoming gradient batches are appended to G without orthonormalization; the Fisher diagonal is used at its raw scale. With 197 chunks accumulated per forward pass, the gradient columns of G and the entries of F are expected to reach extreme magnitudes, inflating the condition number of the projection kernel and driving the matrix inversion to `NaN`. This condition is the negative control that demonstrates the pathology concretely.

CONDITION 2 — GRADIENT ORTHONORMALIZATION ONLY. Each incoming gradient batch is orthonormalized via QR decomposition before insertion into G; the Fisher diagonal is used at its raw scale. This isolates whether gradient-side conditioning alone is sufficient, or whether Fisher-side scaling is also required.

CONDITION 3 — FULL NORMALIZATION (PROPOSED). Gradient batches are QR-orthonormalized at insertion, and the Fisher terms are rescaled by the maximum entry of the ambient metric F^* within each projection computation. This is the proposed scheme; all other hypernetwork experiments use it.

In addition to accuracy and BWT, the condition number $\kappa(A) = \lambda_{\max}(A)/\lambda_{\min}(A)$ of the projection kernel is logged before each inversion as the primary diagnostic. A large $\kappa(A)$ indicates that A is nearly singular: small perturbations in A are amplified by a factor of $\kappa(A)$ in the computed A^{-1}, causing the projected gradient to collapse to zero or `NaN`. A condition number that diverges in Condition 1 but remains bounded in Condition 3 constitutes direct numerical evidence for the pathology and its resolution. All conditions use the Benchmark 2 hypernetwork settings, so any performance difference is attributable to normalization alone.

4.6.5 *Sub-RQ3 — Does the Combined Fisher Metric Yield Consistent Retention Improvements*

BENCHMARK SELECTION AND RATIONALE. Sub-RQ3 compares `iFOPNG` against `FOPNG` across the four standalone benchmarks evaluated by Garg et al. (2026): Permuted-MNIST (5 tasks), Split-MNIST single-head, Split-CIFAR10 multi-head, and Split-CIFAR100 multi-head. All hyperparameters are taken directly from Garg et al.'s Table 1, with the sole exception of the first-task learning rate for MNIST-based benchmarks, where the published value produces a degenerate initialisation; the necessary correction is established empirically in Appendix D (Figure 11). This replication-faithful design ensures that any observed difference between `iFOPNG` and `FOPNG` is attributable to the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ rather than to differences in training conditions.

Split-MNIST multi-head and the two hypernetwork configurations (Split-MNIST HN and Split-CIFAR10 HN) were already conducted for Sub-RQ1 and are included here for completeness; they are not part of the Garg et al. replication and carry no additional inferential weight for Sub-RQ3.

4.6.6 *Sub-RQ4 — Fisher Accumulation Operator*

EXPERIMENTAL DESIGN. `iFOPNG` is run under two accumulation operators, all other settings held constant:

MAXIMUM (PROPOSED). $F_{\text{old}} \leftarrow \max(F_{\text{old}}, F_{\text{new}})$ — non-decaying.

EXPONENTIAL MOVING AVERAGE. $F_{\text{old}} \leftarrow (1-\alpha)\, F_{\text{old}} + \alpha\, F_{\text{new}}$ — decaying, allowing historical importance to fade.

The comparison is run on the longest available task sequences — Permuted-MNIST — since the plasticity cost of monotonic accumulation, if present, only becomes observable as the sequence length grows. Per-task accuracy is tracked across the full sequence rather than reported only as a final aggregate: the diagnostic signature of plasticity paralysis is a progressive decline in *newly learned* task accuracy ($a_{t,t}$ falling as t increases), distinct from forgetting, which appears as decline in *previously learned* task accuracy. If the maximum operator shows superior early-task retention but degrading $a_{t,t}$ in late tasks, while the moving average shows the opposite trade, the two failure modes are cleanly separated and Sub-RQ4 is answered in full.

5 RESULTS

For each sub-research question, we describe the experimental scope, present the key quantitative findings, and highlight the most important patterns. Average accuracy (A_T) is the primary evaluation metric, and backward transfer (BWT) is reported as a secondary measure of forgetting. Standalone target-network results that establish the reference behaviour of projection methods in their original, non-hypernetwork setting are provided in Appendix B (Figure 7).

5.1 *Sub-RQ1: Constructive Synergy or Architectural Conflict?*

Three hypernetwork regimes are evaluated: a standard-capacity Split-CIFAR10 network, an extremely compressed Split-MNIST network (suffocated regime, $d_h = 8$), and a ten-task Split-CIFAR100 network.

STANDARD HYPERNETWORK — SPLIT-CIFAR10. Figure 1 presents average accuracy and BWT for all eight methods on Split-CIFAR10 with a standard-capacity hypernetwork and full normalization enabled. Plain `SGD` at 84.1% establishes the unconstrained reference, and `Adam` at 90.4% defines the upper frontier. `iFOPNG` leads all projection methods at 86.0%, the only one to surpass `SGD`. The remaining methods fall below the `SGD` reference: `EWC` at 83.1%, `FNG` at 81.1%, `OGD` at 79.6%, `FOPNG` at 77.9%, and `ONG` at 71.4%. `Adam` has effectively zero backward transfer (BWT ≈ 0.000), while all projection methods exhibit small but nonzero forgetting (`iFOPNG`: -0.013; `FOPNG`: -0.015).

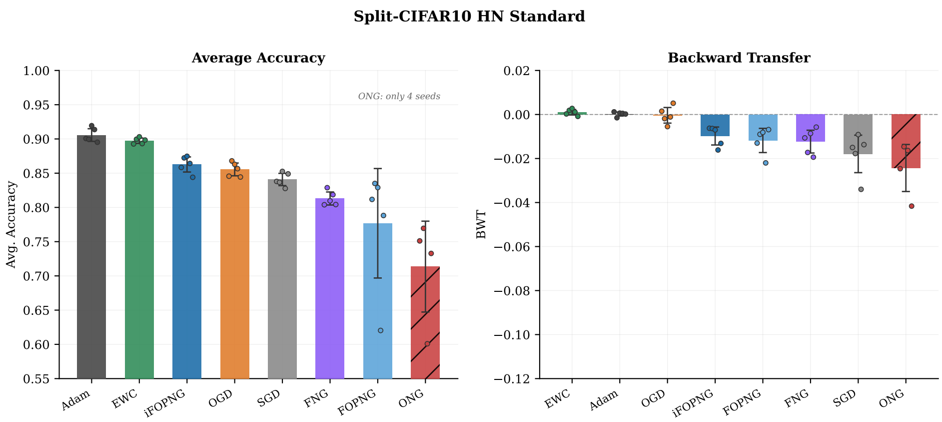

Figure 1: **Split-CIFAR10 hypernetwork (5 tasks, 50 epochs per task, AdamW first-task initialisation at 10^{-3}).** Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. Methods are sorted by average accuracy on the left panel. `Adam` defines the projection-free upper frontier; `SGD` defines the unconstrained reference at 84.1%. `iFOPNG` is the only projection method to exceed `SGD`; the remaining projection variants fall below it. All experiments use full normalization (Section 5.2).

SUFFOCATED REGIME — SPLIT-MNIST, $d_h = 8$. Figure 2 presents the full method comparison. `Adam` leads at 92.7%. `EWC` is the strongest regularization baseline at 87.5%, followed by the Fisher-free projection method `OGD` at 86.1% and `SGD` at 78.4%. The Fisher-based methods fall below: `iFOPNG` at 77.2%, `FOPNG` at 65.4%, and `FNG` at 63.4%. Unlike previous observations, `iFOPNG`, `FOPNG`, and `FNG` all suffer from substantial negative backward transfer (BWT between −0.15 and −0.21), while `Adam` maintains a much smaller negative BWT (−0.068) and `SGD` forgets substantially (−0.180).

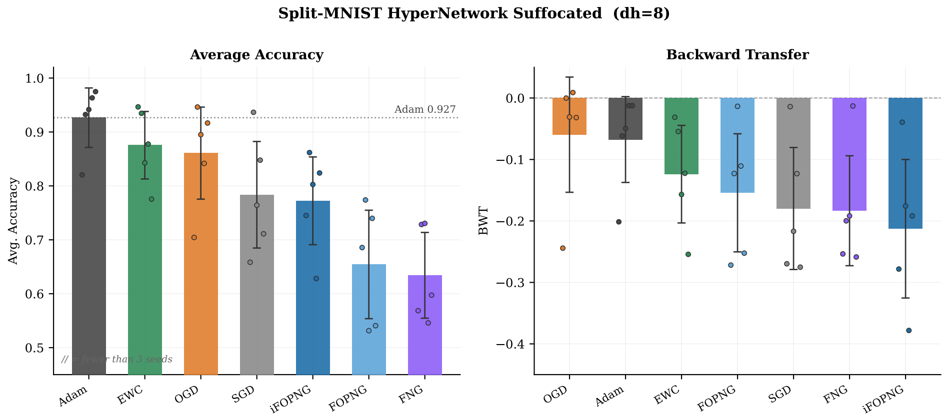

Figure 2: **Split-MNIST SH hypernetwork, suffocated regime ($d_h = 8$, 5 tasks, 15 epochs per task, `AdamW` first-task initialisation).** Left: average accuracy (bars show mean over seeds; circles show individual seeds). Right: backward transfer. The dotted line marks `Adam`'s mean accuracy (92.7%). Fisher-based projection methods (`iFOPNG`, `FOPNG`, `FNG`) suffer from significant negative BWT and lower accuracy than `Adam`, revealing a retention failure in the extreme compression regime. `EWC` is the best-performing baseline at 87.5%, while the Fisher-free `OGD` achieves 86.1%.

EXTENDED TASK COUNT — SPLIT-CIFAR100. At ten tasks on Split-CIFAR100 (Appendix Figure 10), `EWC` unexpectedly leads at 45.5%, with `Adam` second at 42.0% and `iFOPNG` third at 37.8% (random baseline: 10.0%).

5.2 *Sub-RQ2: Normalization Resolves the Gradient Pathology*

This subsection reports the three-condition normalization ablation. Figure 3 traces the $\log_{10}$ condition number of the projection kernel $G^\top FG$ over training steps; Figure 4 shows the resulting average accuracy and BWT under each condition.

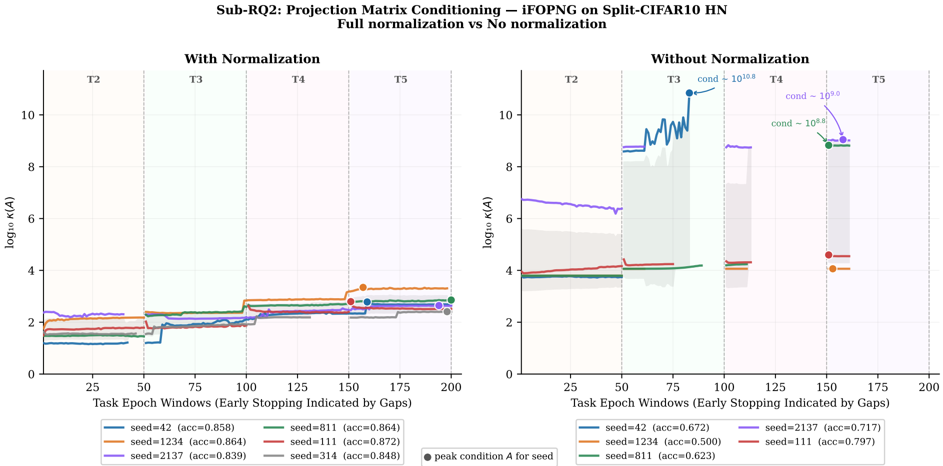

Figure 3: **Projection kernel condition number over training — `iFOPNG` on Split-CIFAR10 HN.** Each line is one seed (legend shows final average accuracy). Task boundaries are marked by vertical dashed lines; gaps indicate early stopping. Left (full normalization): $\log_{10} \kappa(A)$ remains bounded within $[10^{1.3}, 10^{3.3}]$ throughout. Right (no normalization): condition number spikes to $10^{10.8}$ at the T2→T3 transition, causing projection failure. Diamond markers indicate end-of-task condition number values.

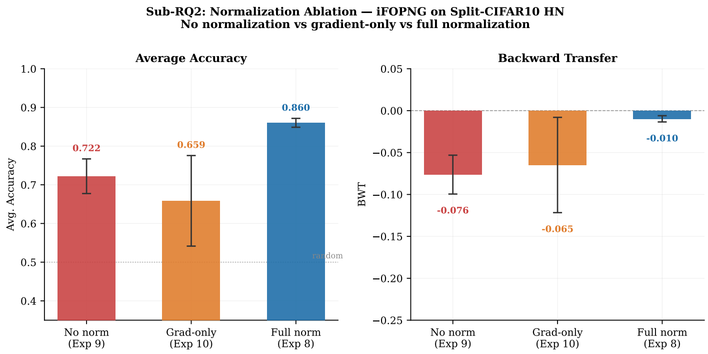

Figure 4: **Normalization ablation — `iFOPNG` on Split-CIFAR10 HN (Sub-RQ2).** Three conditions: no normalization, gradient-only normalization, and full normalization. Left: average accuracy with mean annotated on each bar. Right: backward transfer. Circles show individual seeds; bars show the mean. The dotted line on the left panel marks random-chance performance (50% for binary per-task classification).

WITHOUT NORMALIZATION. The right panel of Figure 3 shows the condition number reaching $10^{10.8}$ at the T2→T3 task transition for the most-affected seed, stabilizing near $10^{8.8}$–$10^{9.0}$ for two others. The result is 72.2% average accuracy and BWT $= -0.076$.

GRADIENT-ONLY NORMALIZATION. Applying unit-ℓ_2 column normalization to G alone recovers 65.9% (BWT -0.065), below both full normalization (86.0%) and the no-normalization baseline (72.2%) (Figure 4).

FULL NORMALIZATION. Combining column-wise ℓ_2 normalization of G with absolute-maximum scaling of F maintains $\log_{10} \kappa(A)$ bounded within $[10^{1.3}, 10^{3.3}]$ across all seeds and task transitions, a reduction of roughly seven orders of magnitude relative to the no-normalization condition (Figure 3). Average accuracy recovers to 86.0% and BWT improves to -0.010 (Figure 4).

5.3 *Sub-RQ3: Parameter Inertia via the Combined Fisher Metric*

Figure 5 presents mean average accuracy for `iFOPNG` and `FOPNG` across all eight benchmarks.

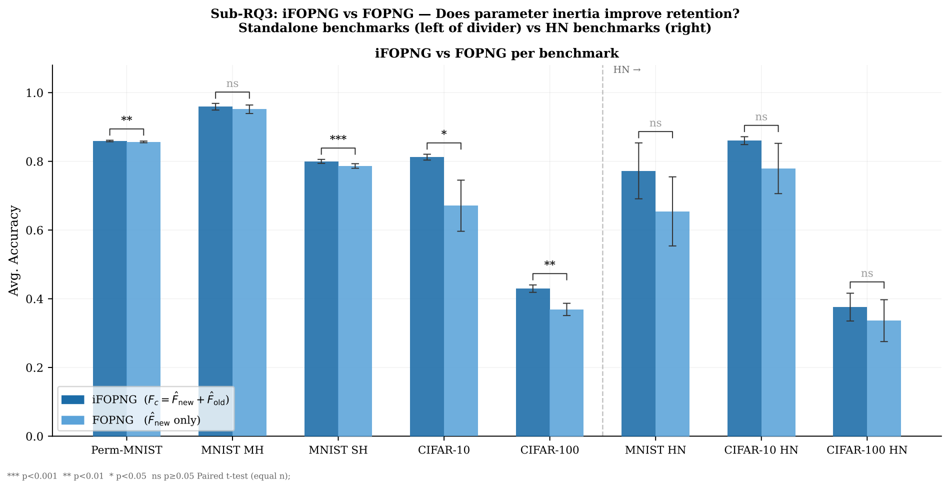

Figure 5: **iFOPNG vs. FOPNG across eight benchmarks (Sub-RQ3).** Grouped bars show mean average accuracy; circles show individual seeds; error bars show ± 1 standard deviation. The dashed vertical line separates standalone (left) from hypernetwork (right) settings. `iFOPNG` uses the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$; `FOPNG` uses only F_{new}.

On the two simplest benchmarks the absolute differences are small, but the two cases diverge statistically. On Permuted-MNIST the 0.3 pp advantage is negligible in size yet highly consistent across seeds, and reaches significance (85.9% vs. 85.7%, $t(4) = 6.38$, $p = .003$, $d_z = 2.85$, $n = 5$). On Split-MNIST multi-head a similarly small near-ceiling gap does not reach significance (95.9% vs. 95.2%, $t(4) = 1.12$, $p = .325$, $n = 5$).

5.4 *Sub-RQ4: Fisher Accumulation Operator*

This subsection compares two strategies for accumulating F_{old} across tasks under the `iFOPNG` combined Fisher metric: the element-wise maximum operator (MAX) and an exponential moving average (EMA). Both are evaluated on a twenty-task Permuted-MNIST sequence with five random seeds. Figure 6 presents the average accuracy and backward transfer trajectories over all twenty tasks.

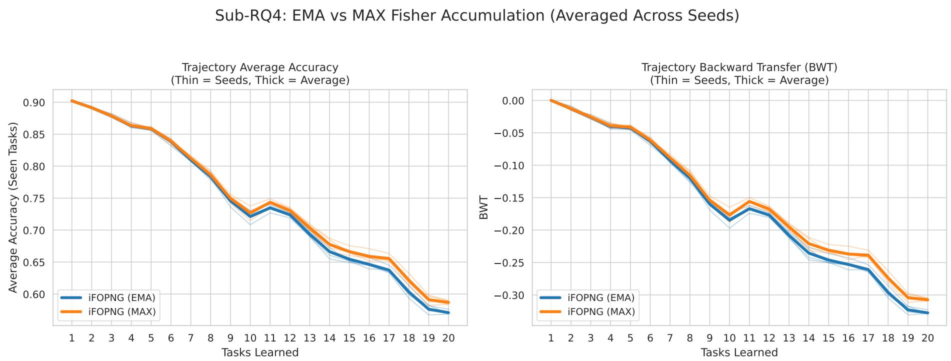

Figure 6: **EMA vs. MAX Fisher accumulation trajectories on 20-task Permuted-MNIST (Sub-RQ4),** $n = 5$ **seeds.** Thick lines show the seed-averaged trajectory; thin lines show individual seeds. Left: average accuracy over tasks seen so far, evaluated after each task is trained. Right: backward transfer over the same sequence. MAX leads EMA by a consistent margin across all five seeds; both strategies produce closely tracking trajectories throughout the full twenty-task sequence.

A paired t-test on final average accuracy establishes MAX as the superior operator: $t(4) = 14.43$, $p < 0.001$, $d_z = 6.45$. MAX leads EMA by 1.50–1.65 percentage points across every seed pair without exception.

6 DISCUSSION

6.1 *Sub-RQ1: Constructive Synergy or Architectural Conflict?*

Projection methods do not outperform a well-tuned regularized hypernetwork optimized via `Adam`. The gap of 4.4 pp at task 5 in favor of `Adam` is consistent across seeds and widens substantially in the suffocated regime ($d_h = 8$, Split-MNIST SH), where it reaches 15–30 pp.

Critically, the BWT data reveal that in the suffocated regime this is a *retention* failure rather than a pure plasticity failure: Fisher-based projection methods suffer from substantial negative backward transfer (BWT between -0.15 and -0.21) alongside collapsed accuracy. This indicates that under extreme parameter bottlenecks, the curvature-weighted constraint fails to protect prior knowledge, permitting catastrophic interference. `OGD`'s relative resilience at $d_h = 8$ (BWT ≈ -0.06) is consistent with this reading: without the volatile Fisher weighting, its simpler subspace constraint ironically provides a more stable anchor against forgetting. This extreme compression setting can be read as a proxy for the tight parameter budgets of edge-device deployment, suggesting that the retention collapse of Fisher-based methods under such budgets could limit their practical utility.

We attribute the broader gap to a fundamental architectural incompatibility. Projection methods operate by modifying the raw gradient before the optimizer state update, which means the first-order momentum estimate and the adaptive scaling factor maintained by Adam are effectively discarded at each projected step. In the highly nonlinear, compressed weight space of a hypernetwork meta-generator, momentum is not merely a convergence accelerant — it is a structural regularizer that prevents the optimizer from committing prematurely to poorly-conditioned gradient directions. Re-initializing the optimizer's internal state at each step destabilizes the generative manifold and reduces the quality of the generated weights, particularly under extreme bottleneck compression. This finding is consistent with Hu, Yu, Cheng, Liu, and Song (2026), who demonstrate that gradient modification methods interact adversely with Adam's internal state under certain curvature regimes, and aligns with the broader observation that projection-based continual learning methods were originally designed for SGD-like updates.

The ten-task Split-CIFAR100 result introduces a partial rank reversal that complicates the Sub-RQ1 conclusion: `EWC` unexpectedly leads at 45.5%, overtaking both `Adam` (42.0%) and `iFOPNG` (37.8%). As the number of tasks grows, the gradient subspace spanned by G becomes an increasingly coarse approximation to the full historical interference manifold, while the functional regularizer grows more informative with each additional stored weight snapshot. Whether this effect persists at the twenty-task scale, and whether an expanded memory budget would recover the projection advantage, remains to be investigated.

6.2 *Sub-RQ2: Projection-Scoped Normalization Resolves the Pathology*

The pathology is purely numerical rather than conceptual: the intersection of chunked hypernetworks and Fisher-weighted gradient projection is geometrically coherent but arithmetically fragile without explicit local rescaling. The projection kernels of `FOPNG` and `iFOPNG` share the same algebraic structure — $A = G^\top F_{\text{old}}\, M^{-1}\, F_{\text{old}}\, G + \lambda I$ — so the chunk-induced inflation of F_{old} degrades $\kappa(A)$ identically in both cases, and the normalization fix generalises across both methods. `OGD` and `ONG` are structurally immune: their kernel $A = G^\top G + \lambda I$ contains no Fisher weighting, so gradient orthonormalization alone suffices.

The gradient-only condition confirms that joint rescaling is necessary rather than merely sufficient: column-normalising G alone worsens the scale disparity between the unit-norm gradient basis and the still chunk-inflated F_{old}, yielding 65.9% — below the no-normalization baseline of 72.2%. A well-conditioned kernel requires the two components to be co-scaled within the same local algebra.

Severity scales with chunk count C: gradient magnitudes accumulate as $C \cdot g_{\text{unit}}$, Fisher entries as $C^2 \cdot F_{\text{unit}}$, and the unnormalized kernel inflates by order C^4 per task. The 72.2% no-normalization result is a relatively benign failure specific to Split-CIFAR10's moderate chunk count; in larger configurations normalization absence produces categorical failure — runs crash or converge to random-chance accuracy. The elevated failure rate on Split-CIFAR100 hypernetwork is consistent with this scaling argument even after full normalization is applied, suggesting that chunk-Jacobian-aware rescaling may be warranted at larger scales.

6.3 *Sub-RQ3: Parameter Inertia via the Combined Fisher Metric*

`iFOPNG` outperforms `FOPNG` in all eight settings, confirming that the combined metric $F_c = F_{\text{new}} + F_{\text{old}}$ yields a consistent *direction* of retention improvement. The gap is positive in every case; its statistical reliability, however, is governed by inter-seed consistency rather than by raw magnitude. The advantage is significant on four of the five standalone benchmarks: Permuted-MNIST (+0.3 pp, $p = .003$, $d_z = 2.85$), Split-MNIST single-head (+1.3 pp, $p < 0.001$, $d_z = 10.76$), Split-CIFAR10 (+14.1 pp, $p = .024$), and Split-CIFAR100 (+6.0 pp, $p = .003$). Permuted-MNIST is instructive precisely because its magnitude is negligible: a 0.3 pp gap is statistically reliable only because the per-seed difference is almost invariant, so significance here reflects consistency, not practical importance. The single standalone exception is Split-MNIST multi-head (+0.7 pp, $p = .325$), a near-ceiling benchmark whose separate per-task heads leave little room for any method to separate. The largest practical gains arise on the harder standalone benchmarks and across the three hypernetwork settings, but there the elevated inter-seed variance prevents significance, so the hypernetwork evidence is suggestive rather than conclusive. Direction is uniform across all eight settings while magnitude and significance vary independently, so Sub-RQ3 is answered affirmatively for the standalone setting and tentatively for the hypernetwork setting.

iFOPNG does not close the gap to Adam, confirming that the inertial combined metric improves retention by weighting historically important coordinates more heavily, but cannot recover the adaptive momentum that Adam leverages for efficient navigation of the generative manifold.

6.4 *Sub-RQ4: Fisher Accumulation Operator*

On a twenty-task Permuted-MNIST sequence, the element-wise maximum (MAX) accumulation strategy systematically outperformed the exponential moving average (EMA). While the 1.60 pp advantage is small, it is statistically unambiguous ($p < 0.001$).

Crucially, the hypothesized plasticity cost of MAX's monotonic growth did not materialize. Because our normalization scheme rescales both Fisher terms, the projection becomes insensitive to absolute scale, relying solely on relative parameter distributions. Therefore, MAX remains the definitive practical choice: it is simpler, hyperparameter-free, and marginally superior, though highly heterogeneous task distributions could potentially introduce divergence.

6.5 *Limitations*

Several limitations should be acknowledged.

First, the hypernetwork results presented here are limited in task-sequence length: Split-MNIST and Split-CIFAR10 use five tasks, and only the preliminary Split-CIFAR100 result extends to ten. Longer task sequences may produce different relative orderings: in particular, the advantage of hard projection constraints over soft functional regularization may grow with sequence length, as soft penalties are known to suffer from compounding output drift (Farquhar & Gal, 2018), while the projection subspace scales with the gradient budget rather than the task count. The preliminary ten-task Split-CIFAR100 result, in which EWC overtakes both Adam and iFOPNG, is suggestive but insufficient to establish a trend; a systematic sweep over task count is warranted.

Second, the results are specific to the constrained-capacity regimes examined here. For Split-MNIST, a standard-capacity hypernetwork solves the benchmark near-trivially regardless of the continual learning method applied; the suffocated setting ($d_h = 8$) is deliberately chosen to place the architecture below saturation, where method differences are measurable. For Split-CIFAR10, the convolutional target network is large enough that the hypernetwork already operates under a natural capacity constraint, and meaningful method separation emerges without artificial bottlenecking. Whether the projection–hypernetwork interaction studied here generalises to harder benchmarks requiring larger target architectures — where the hypernetwork must generate weights for deeper convolutional or attention-based networks and the task itself is sufficiently complex to prevent ceiling effects — remains an open question. In such settings, the gradient accumulation pathology documented here would be substantially more severe: chunk counts scale with the target parameter count, so the condition number blowup $\sim C^4$ identified here represents a lower bound on the numerical challenge at scale.

Third, the proposed normalization scheme—QR orthonormalization combined with absolute-maximum Fisher scaling—is fundamentally heuristic in nature. While it successfully eliminates the numerical pathology of condition number collapse, it applies isotropic rescaling to the projection basis. This likely discards important geometric properties of the Fisher metric, temporarily divorcing the subspace constraint from the true curvature of the parameter space. A mathematically principled formulation that stabilizes the projection algebra while strictly preserving the theoretical guarantees of Fisher-based methods remains an open question for future work.

Fourth, the hypernetwork experiments do not use the hyperfan initialisation of Chang, Flokas, and Lipson (2020) adopted by Oswald et al. (2020), which is designed to place the *generated* target weights in a well-conditioned variance regime at the start of training. We instead seed the first task with `AdamW` (Section 5.1). Because the projection methods bootstrap their gradient memory G and Fisher estimate F from the first-task solution, the quality of that initial basis is consequential: as the seed-2137 failure (Figure 15) demonstrates, a degenerate first-task solution permanently forecloses all subsequent projection. A principled hyperfan initialisation could yield a cleaner and more stable initial projection subspace, and we cannot rule out that some portion of the projection–baseline gap reported here reflects our initialisation choice rather than the projection geometry itself. Quantifying this is left to future work.

Fifth, all hypernetwork experiments use a uniform learning rate of 10^{-3} and a fixed regularization strength β across all methods, a deliberate choice to isolate projection geometry as the sole independent variable (Section 4.6). This design trades internal comparability for external generalisability: method-specific hyperparameter tuning—particularly per-method learning rates and λ damping coefficients—could plausibly shift the relative rankings observed here. In the standalone setting the published values of Garg et al. (2026) are used directly, so that comparison carries stronger external validity. The hypernetwork results should therefore be read as establishing the *structure* of the projection–hypernetwork interaction under a controlled configuration, not as upper-bound performance estimates for any individual method.

6.6 *Directions for Future Work*

The findings of this thesis highlight several avenues for future research. The most immediate extension is developing projection-compatible adaptive optimizers that maintain momentum across constrained updates (Hu et al., 2026).

Additionally, the heuristic QR normalization and FIM max scaling scheme must be replaced with a principled reformulation that integrates chunk-Jacobian curvature directly into the projection algebra.

Finally, future work should systematically evaluate these interactions on extended task sequences, where soft regularization penalties may degrade, and on larger target architectures, where the documented condition number blowup will require more robust mitigation.

7 CONCLUSION

This thesis investigated integrating gradient projection methods into chunked hypernetworks for continual learning. We primarily diagnosed the architectural boundaries and numerical pathologies governing this core integration. Alongside these fundamental structural findings, we also evaluated adjacent methodological optimizations regarding parameter inertia and Fisher matrix accumulation.

SUB-RQ1 — PARTIAL SYNERGY AND THE MOMENTUM BOTTLENECK. The synergy between hard gradient projection and soft functional regularization is constructive but bounded. While `iFOPNG` outperforms unconstrained `SGD` in standard-capacity settings, standalone regularized hypernetworks optimized via `Adam` remain vastly superior.

This performance gap stems from an architectural conflict, not projection geometry. Under extreme compression ($d_h = 8$), Fisher-based projection methods suffer a severe *retention* failure, experiencing substantial negative backward transfer as the constraint fails to protect prior knowledge. Because projection modifies raw gradients before the optimizer updates, it discards adaptive momentum. In hypernetworks, this momentum acts as a crucial structural regularizer; without it, the network commits prematurely to poorly-conditioned gradient directions, destabilizing the generative manifold and causing catastrophic interference.

Ultimately, the loss of adaptive momentum is the binding constraint. Integrating gradient projection into hypernetworks will require projection-compatible adaptive optimizers, rather than post-hoc gradient modifications (Hu et al., 2026).

SUB-RQ2 — NORMALIZATION IS A NECESSARY CONDITION. Without explicit local rescaling, projection methods are numerically incompatible with chunked hypernetworks: matrix inversion in the projection kernel degrades or collapses entirely depending on chunk count, pulling performance below plain SGD. QR orthonormalization of the gradient basis combined with absolute-maximum Fisher scaling resolves the pathology in the tested configurations. This normalization scheme is a necessary engineering precondition for any future combination of gradient projection with chunked meta-learners, regardless of which projection algebra is employed.

SUB-RQ3 — PARAMETER INERTIA VIA COMBINED FISHER METRIC. Replacing the task-specific Fisher metric with the combined inertial metric (F_c) improves retention in direction across every benchmark tested. The advantage is statistically significant on four of the five standalone benchmarks, including Permuted-MNIST, where it is negligible in size (+0.3 pp) yet reliable on account of its consistency across seeds; the only standalone null is the near-ceiling Split-MNIST multi-head (+0.7 pp, $p = .325$). The advantage is largest on Split-CIFAR10 (+14.1 pp, $p = .024$), alongside a substantial gain on Split-MNIST single-head (+1.3 pp). The same positive direction carries into the three hypernetwork settings, though none reaches significance under the elevated inter-seed variance of that regime. Integrating parameter inertia therefore yields a positive retention advantage in every setting: statistically robust across the standalone benchmarks, and directionally consistent but underpowered in the hypernetwork settings.

SUB-RQ4 — MAX ACCUMULATION IS THE CLEAR PRACTICAL CHOICE. On a twenty-task sequence the element-wise maximum operator was statistically superior to an exponential moving average ($\sim$1.60 pp, $p < 0.001$), and the hypothesised plasticity cost of its monotonic growth did not materialise. MAX is therefore preferred: hyperparameter-free, simpler, and marginally better.

IMPLICATIONS FOR THE FIELD. The results establish a clear empirical boundary for the current generation of gradient projection methods: they can be made numerically stable in chunked hypernetwork settings via local rescaling, and their Fisher geometry can be improved via the combined Fisher metric F_c, but the momentum incompatibility prevents them from reaching the performance frontier of adaptive unconstrained optimizers in generative parameter spaces. The normalization contribution is directly applicable to any method that maintains a gradient subspace memory alongside a chunked weight generator. The iFOPNG contribution extends naturally to any continual learning setting where Fisher overlap accumulates across tasks. Together, they narrow the gap between projection-based and regularization-based approaches, and provide a principled foundation for the next step: a projected adaptive optimizer that respects both the geometric constraints of the projection and the curvature structure that adaptive moment estimates capture.

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8 APPENDIX

APPENDIX A: IFOPNG THEOREM AND PROOF

Notation

Throughout this proof, $g \in \mathbb{R}^p$ denotes the current gradient, $G \in \mathbb{R}^{p \times m}$ the memory matrix of m historical gradient directions (each orthonormalised at insertion via QR; the proof does not rely on this property), and F_{new}, F_{old} the current-task and accumulated diagonal empirical Fisher tensors, with $F_c = F_{\text{new}} + F_{\text{old}}$. Diagonal Fisher tensors are treated as vectors acting by element-wise multiplication and are identified with the corresponding diagonal matrices, which are symmetric. For a symmetric positive-definite S we write $\|w\|_S = \sqrt{w^\top S w}$. The projection kernel is

$$A := G^\top F_{\text{old}}\, F_c^{-1}\, F_{\text{old}}\, G \;\in\; \mathbb{R}^{m \times m}. \tag{19}$$

Assumption 1 (Regularity). *F_c is positive definite and the kernel A is invertible.*

Both conditions hold unconditionally in the implementation, which inverts the damped matrices $F_c + \hat{\lambda} I$ and $A + \hat{\lambda} I$ with $\hat{\lambda} > 0$ (Section 4.5.1); the theorem below states the exact $\lambda \to 0$ result.

Theorem 1 (iFOPNG Update). *Under Assumption 1, the solution to*

$$\begin{aligned} \min_u \quad & \|g - F_c^{-1} u\|_{F_c}^2 \\ \text{s.t.} \quad & F_{\text{old}}^{-1} u \in \text{Col}(G), \\ & \|v\|_{F_c} = \eta, \end{aligned} \tag{20}$$

with parameter update $v = \gamma\, F_c^{-1}(g - u)$, is given by Equation (15):

$$v^* = -\eta\, \frac{F_c^{-1} P g}{\sqrt{(Pg)^\top F_c^{-1} P g}}, \qquad P = I - F_{\text{old}}\, G\, A^{-1}\, G^\top F_{\text{old}}. \tag{21}$$

Proof. **Step 1 (parameterise the constraint).** The constraint $F_{\text{old}}^{-1} u \in \text{Col}(G)$ means $F_{\text{old}}^{-1} u = G\alpha$ for some $\alpha \in \mathbb{R}^m$, i.e. $u = F_{\text{old}}\, G\, \alpha$. The optimisation over the constrained u thus reduces to an unconstrained optimisation over $\alpha \in \mathbb{R}^m$.

Step 2 (expand the objective). Writing $\|w\|_{F_c}^2 = w^\top F_c\, w$ and using $F_c^{-1} F_c = I$,

$$\|g - F_c^{-1} u\|_{F_c}^2 = g^\top F_c\, g \;-\; 2\, g^\top u \;+\; u^\top F_c^{-1} u. \tag{22}$$

Substituting $u = F_{\text{old}}\, G\, \alpha$, the linear term becomes $-2\,(G^\top F_{\text{old}}\, g)^\top \alpha$ and the quadratic term becomes $\alpha^\top A\, \alpha$, since $G^\top F_{\text{old}}\, F_c^{-1}\, F_{\text{old}}\, G = A$. The first term $g^\top F_c\, g$ does not depend on α.

Step 3 (minimise). The reduced objective is quadratic in α with Hessian $2A \succ 0$, hence strictly convex. Setting its gradient to zero gives the normal equations $A\,\alpha^* = G^\top F_{\text{old}}\, g$, with unique solution

$$\alpha^* = A^{-1}\, G^\top F_{\text{old}}\, g. \tag{23}$$

Step 4 (back-substitute). Inserting α^* into $u = F_{\text{old}}\, G\, \alpha$ gives $u^* = F_{\text{old}}\, G\, A^{-1}\, G^\top F_{\text{old}}\, g$, so

$$g - u^* = \left(I - F_{\text{old}}\, G\, A^{-1}\, G^\top F_{\text{old}}\right) g = Pg. \tag{24}$$

Step 5 (apply the trust region). For $v = \gamma\, F_c^{-1} Pg$,

$$\|v\|_{F_c}^2 = \gamma^2\,(Pg)^\top F_c^{-1} F_c\, F_c^{-1}(Pg) = \gamma^2\,(Pg)^\top F_c^{-1}(Pg). \tag{25}$$

Imposing $\|v\|_{F_c} = \eta$ fixes $|\gamma| = \eta / \sqrt{(Pg)^\top F_c^{-1} Pg}$. The negative root is selected so that with empty memory ($P = I$) the update reduces to the natural-gradient descent step under F_c, matching the `FOPNG` convention (Garg et al., 2026). This yields the stated update:

$$v^* = -\eta\, \frac{F_c^{-1} Pg}{\sqrt{(Pg)^\top F_c^{-1} Pg}}. \tag{26}$$

□

Remark (Relationship to FOPNG). The single substitution $F_{\text{new}} \to F_c$ recovers `FOPNG`: the proof is identical with F_c replaced by F_{new} in the metric, the kernel A, and the trust region. `iFOPNG` differs only in that the ambient metric carries the accumulated historical curvature F_{old} in addition to the current-task curvature.

Remark (Inertia mechanism). The trust-region budget splits additively across the two geometries:

$$\|v\|_{F_c}^2 = v^\top F_{\text{new}}\, v + v^\top F_{\text{old}}\, v = \|v\|_{F_{\text{new}}}^2 + \|v\|_{F_{\text{old}}}^2 = \eta^2. \tag{27}$$

A coordinate with large $(F_{\text{old}})_i$ consumes more of the shared budget η^2 per unit of motion and is therefore moved less. This is the sense in which historical importance acts as inertia rather than as a restoring force: no reference point and no penalty term enter the objective, in contrast to EWC (Kirkpatrick et al., 2017).

Algorithm Statements

Algorithm 2 is reproduced from Garg et al. (2026) without modification. It pre-multiplies each stored gradient by the Fisher estimate of the task it was collected on, absorbing the metric weighting into $\tilde{G}$ at collection time so that $\tilde{G}$ approximates $F_{\text{old}}\, G$ (up to subsequent element-wise maximum updates of F_{old}) and the per-iteration projection reduces to a single matrix–vector product against $\tilde{G}$. Empirically the two algorithms produce indistinguishable results on all benchmarks evaluated here; `preIFOPNG` is the implementation preferred in practice due to its reduced cost.

Algorithm 1 iFOPNG Algorithm (extends `FOPNG` (Garg et al., 2026) via combined metric F_c)

Require: Initial parameters θ_0, task sequence $\{D_1, \ldots, D_T\}$, learning rate η, damping λ, number of stored gradients per task k, number of epochs E

Initialize gradient memory $G \leftarrow [\,]$. Train on task D_1 using Adam or SGD.
Compute $F_{\text{old}} \leftarrow F(D_1)$; Store $G \leftarrow [g_1, \ldots, g_k]$
for task $t = 2, 3, \ldots, T$ **do**
    for epoch $e = 1, \ldots, E$ **do**
        Recompute new-task Fisher: $F_{\text{new}} \leftarrow F(D_t)$;
        Form combined metric: $F_c \leftarrow F_{\text{new}} + F_{\text{old}}$
        for all batches (x, y) in D_t **do**
            Compute gradient $g \leftarrow \nabla_\theta \mathcal{L}_t(\theta)$
            Compute v^* according to Theorem 1 under metric F_c
            Update parameters: $\theta \leftarrow \theta + v^*$
        end for
    end for
    Store gradients: $G \leftarrow [G,\ g_{k(t-1)+1}, \ldots, g_{kt}]$
    Update accumulated Fisher: $F_{\text{old}} \leftarrow \max(F_{\text{old}},\ F_{\text{new}})$
end for

Algorithm 2 preIFOPNG Algorithm (Garg et al., 2026), adapted for the iFOPNG combined metric

Require: Initial parameters θ_0, task sequence $\{D_1, \ldots, D_T\}$, learning rate η, damping λ, number of stored gradients per task k, number of epochs E

```
Initialize pre-Fisher memory G̃ ← [ ]. Train on task D_1 using Adam or SGD.
Compute F_1 ← F(D_1);   F_old ← F_1;   Store G̃ ← [F_1 ⊙ g_1, ..., F_1 ⊙ g_k]
for task t = 2, 3, ..., T do
    for epoch e = 1, ..., E do
        Recompute new-task Fisher: F_new ← F(D_t);
        Form combined metric: F_c ← F_new + F_old
        for all batches (x, y) in D_t do
            Compute gradient g ← ∇_θ L_t(θ)
            Compute v* according to Theorem 1 using G̃ in place of G
            Update parameters: θ ← θ + v*
        end for
    end for
    Compute F_t ← F(D_t);   Store G̃ ← [G̃, F_t ⊙ g_{k(t−1)+1}, ..., F_t ⊙ g_{kt}]
    Update accumulated Fisher: F_old ← max(F_old, F_new)
end for
```

APPENDIX B: STANDALONE TARGET-NETWORK RESULTS

This appendix presents the full method comparison on standalone target networks — configurations where a conventional MLP is trained directly via gradient projection, without any hypernetwork wrapper or functional regulariser. These results establish the baseline behaviour of each projection method in its intended setting, prior to the hypernetwork integration examined in the main text. All five benchmark streams are reported as per-task accuracy trajectories in a single figure (Figure 7): Permuted-MNIST (5 tasks), Split-MNIST single-head (5 tasks), Split-MNIST multi-head (5 tasks), Split-CIFAR10 (5 tasks), and Split-CIFAR100 (10 tasks).

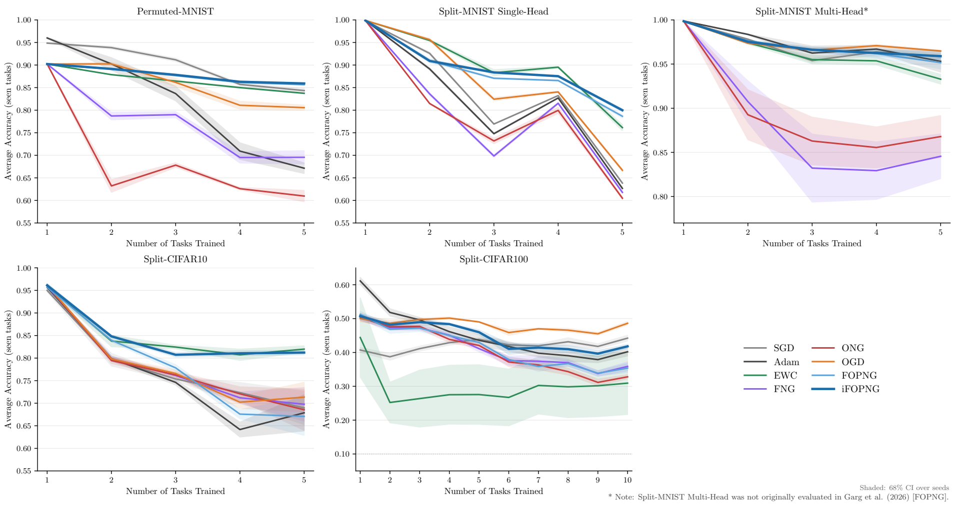

Figure 7: **Standalone target-network accuracy trajectories across all benchmarks, replicating the evaluation protocol of Garg et al. (2026).** Each panel is one benchmark: Permuted-MNIST (5T), Split-MNIST single-head (5T), Split-MNIST multi-head (5T), Split-CIFAR10 (5T), and Split-CIFAR100 (10T). Within each panel, the x-axis is the number of tasks trained so far and the y-axis is the average accuracy over all tasks seen up to that point. Lines show the seed mean for each method; shaded bands show the 68% confidence interval over seeds. The trajectory format is used because the per-task curves are directly comparable to those reported by Garg et al. (2026), whose protocol this replicates. Backward-transfer (BWT) values quoted in the commentary below are reported summary statistics; this figure shows accuracy only.

Per-benchmark commentary follows. All accuracies are seed means after the final task, and BWT is the seed-mean backward transfer over the canonical five seeds. *Permuted-MNIST (5T).* `iFOPNG` (85.9%, BWT $= -0.04$) and `FOPNG` (85.7%, BWT $= -0.05$) trace the flattest trajectories, retaining near-peak accuracy across all five tasks, with `SGD` (84.3%) and `EWC` (83.9%, BWT $= -0.03$) close behind. `OGD` (80.6%, BWT $= -0.18$) sits a step lower. `FNG` (69.6%), `Adam` (67.1%), and `ONG` (61.0%) decline steadily as tasks accumulate, with strongly negative BWT (-0.31, -0.37, -0.44 respectively), reflecting severe catastrophic forgetting absent an effective projection or regularisation mechanism. *Split-MNIST SH (5T).* `iFOPNG` maintains the highest trajectory throughout, ending at 80.0% (BWT $= -0.11$), with `FOPNG` close behind at 78.7% (BWT $= -0.14$) — confirming the Sub-RQ3 finding that the inertial combined-metric preconditioner provides a consistent advantage under output-level interference. `EWC` (76.1%) forms a second tier. `OGD` (66.7%, BWT ≈ -0.38) is far weaker than in the multi-head setting, reflecting the harder gradient-subspace overlap of the shared head. The baselines `SGD` (63.8%), `Adam` (62.7%), `FNG` (61.8%), and `ONG` (60.5%) decline steeply across tasks (BWT ≈ -0.42 to -0.48). *Split-MNIST MH (5T).* All trajectories remain near the ceiling, reflecting the low difficulty of the binary per-task splits. `OGD`, `SGD`, `iFOPNG`, `Adam`, and `FOPNG` lead ($\approx$ 96.5–95%), with `EWC` a short step below ($\approx$ 93%); `ONG` (86.8%) and `FNG` (84.6%) are the exceptions, dipping over the sequence with strongly negative BWT that indicates convergence instability rather than gradual forgetting. *Split-CIFAR10 (5T).* `EWC` (81.9%) and `iFOPNG` (81.2%) lead with moderate forgetting (BWT ≈ -0.09), tracing the highest trajectories. The remaining methods fall to 67–71% with substantially higher seed variance (wider CI bands) and stronger decline. `FOPNG` is the weakest projection method at 67.4% — a 13.8 pp gap below `iFOPNG` that motivates the Sub-RQ3 comparison — and `Adam` (68.1%) performs comparably to the weaker projection methods, confirming that the unconstrained baseline is not dominant on harder standalone benchmarks. *Split-CIFAR100 (10T).* `EWC` (54.0%) traces the highest trajectory, followed by `OGD` (49.0%) and `SGD` (44.6%). `iFOPNG` (42.9%) holds a 6.0 pp advantage over `FOPNG` (36.9%), consistent with the Sub-RQ3 finding that parameter inertia provides increasing benefit as task count and Fisher overlap grow. `Adam` (40.4%) sits just behind `iFOPNG`, while `FNG` (37.1%) closely mirrors `FOPNG`. `ONG` declines fastest, ending lowest at 33.8% (BWT ≈ -0.38). Random-chance performance is 10%.

APPENDIX C: HYPERNETWORK ARCHITECTURE ABLATIONS

This appendix documents two architectural ablations that motivated the hypernetwork configurations used in the main experiments. Figure 8 establishes the diagnostic capacity level $d_h = 8$ for the suffocated Split-MNIST regime (Sub-RQ1, Benchmark 1). Figure 9 presents the joint sweep over regularization strength β and chunk size c used to select the operating point for all main hypernetwork experiments.

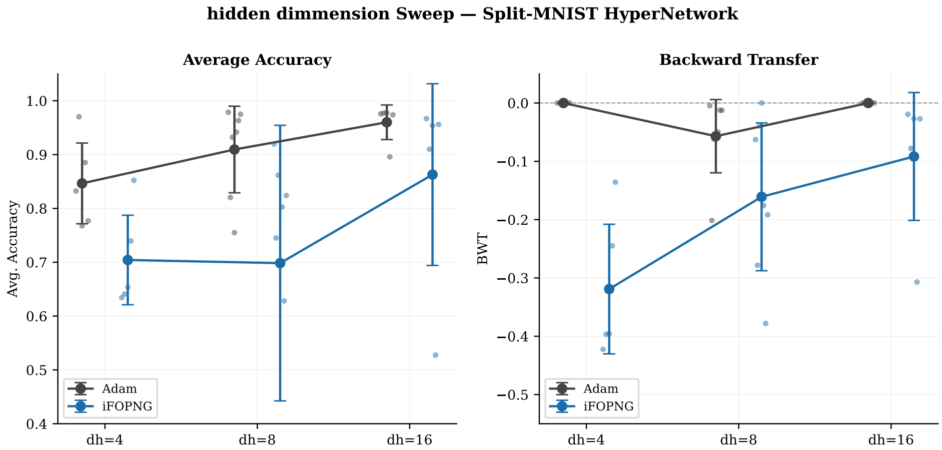

Figure 8: **Hidden-dimension sweep — Split-MNIST SH hypernetwork ($d_h \in \{4, 8, 16\}$, 5 tasks, 15 epochs per task, `AdamW` first-task initialisation at 10^{-3}).** Left: average accuracy; right: backward transfer. Thick markers show seed means; error bars show ± 1 standard deviation; individual seeds are shown as dots. Only `Adam` and `iFOPNG` are plotted to isolate the projection–baseline gap across capacity levels. Three capacity levels are compared. At $d_h = 4$, the Adam–iFOPNG gap is 14.0 pp, but individual seed variance for `iFOPNG` is extreme (range ≈ 0.25), indicating that the bottleneck is too severe for reliable projection convergence. At $d_h = 8$, the gap widens to 23.5 pp and variance contracts substantially: seeds cluster in the range 0.63–0.92 for `Adam` and 0.63–0.75 for `iFOPNG`, making the failure mode severe and consistently replicable. At $d_h = 16$, projection methods partially recover — the gap narrows to 9.4 pp — as the larger gradient budget reduces the relative cost of subspace projection. The backward transfer panel mirrors this pattern: `iFOPNG` BWT improves monotonically with capacity (≈ -0.32 at $d_h = 4$, ≈ -0.17 at $d_h = 8$, ≈ -0.11 at $d_h = 16$), while `Adam` remains near zero throughout. $d_h = 8$ was retained as the primary diagnostic level: the plasticity failure is both severe and consistently replicable, whereas $d_h = 4$ yields near-random results in some seeds and $d_h = 16$ partially obscures the failure mode.

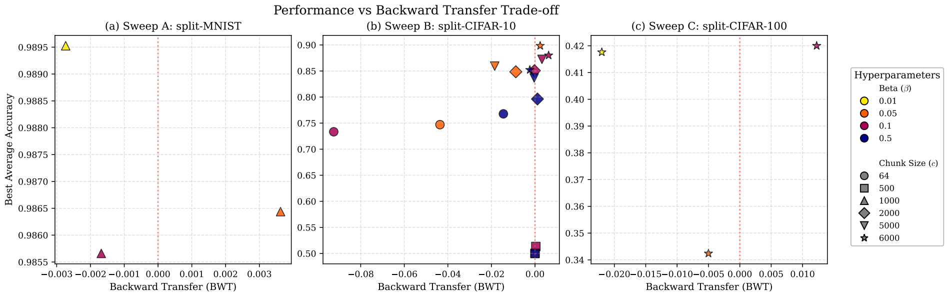

Figure 9: **Beta and chunk-size sweep — hypernetwork performance vs. backward transfer trade-off across three benchmarks.** Each panel plots mean average accuracy (y-axis) against backward transfer (x-axis) for all evaluated combinations of regularization strength $\beta \in \{0.01, 0.05, 0.1, 0.5\}$ and chunk size $c \in \{64, 500, 1000, 2000, 5000, 6000\}$. Marker shape encodes chunk size; marker colour encodes β. The vertical red dashed line marks BWT $= 0$; points to its left exhibit net retention, points to its right exhibit net forgetting. The probe method is `Adam` with the functional regularizer active. Split-MNIST (panel a) is insensitive to both hyperparameters: all combinations achieve 98.55–98.95% accuracy with BWT spanning only ≈0.006, indicating that the task is too simple to discriminate architectures. Split-CIFAR10 (panel b) shows clear structure. Small chunk sizes ($c \in \{64, 500\}$) cluster in the lower-left regardless of β, reaching only 74–75% accuracy; larger chunks progressively shift the Pareto frontier upward. The combination $c = 6000$, $\beta = 0.1$ sits on the efficiency frontier: it achieves 87–88% average accuracy with small negative BWT, providing the best retention–plasticity balance. The combination $c = 6000$, $\beta = 0.5$ yields similar accuracy but with marginally higher forgetting. Split-CIFAR100 (panel c) confirms the selection: $c = 6000$, $\beta = 0.5$ achieves the highest accuracy (≈42%) but at the cost of slightly positive BWT, suggesting the stronger regularizer over-constrains plasticity on a 10-task sequence. $c = 6000$, $\beta = 0.1$ achieves ≈34% with negative BWT and was therefore selected as the operating point for all main hypernetwork experiments, providing consistent Pareto-frontier performance across both CIFAR benchmarks without the positive-BWT artefact observed at higher β.

APPENDIX D: ADDITIONAL EXPERIMENT FIGURES

This appendix collects six figures that are referenced in the Results section but omitted from the main text for space. Figure 10 shows the full method comparison on the ten-task Split-CIFAR100 hypernetwork benchmark. Figure 11 documents the SGD first-task convergence diagnostic. Figure 12 reports the `OGD` learning-rate sensitivity sweep. Figure 13 provides the per-seed paired comparison underlying the Sub-RQ4 statistical test. Figure 14 compares the SVD, FIFO, and STOP gradient-memory compression strategies on 20-task Permuted-MNIST. Figure 15 documents the seed-2137 initialization failure on the Split-MNIST hypernetwork benchmark that motivated the seed-replacement policy of Section 4.6.2.

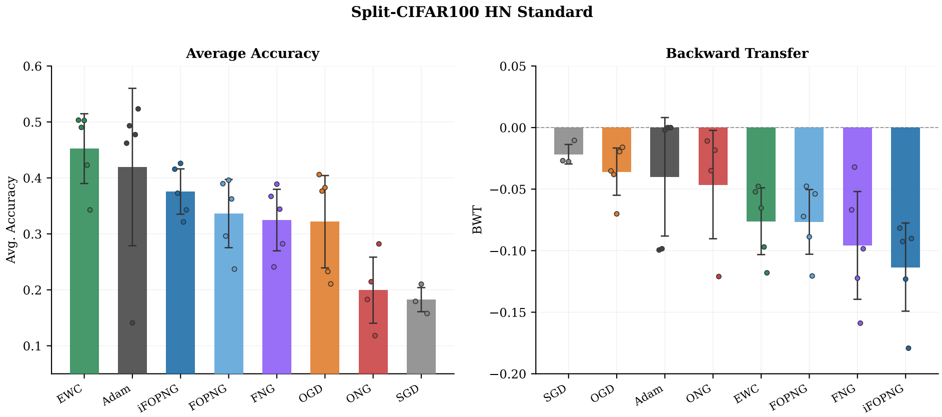

Figure 10: **Split-CIFAR100 hypernetwork — 10 tasks, 50 epochs per task, `AdamW` first-task initialisation at 10^{-3}.** Left: average accuracy. Right: backward transfer. Random-chance performance is 10% (10-way classification per task). `EWC` leads at 45.5%, with `Adam` second at 42.0% and `iFOPNG` third at 37.8% — a rank ordering that inverts the five-task Split-CIFAR10 pattern (where projection methods outperform `EWC` and closely trail `Adam`). This reversal is discussed in the main text; the ten-task sequence length and the higher inter-task similarity of the CIFAR-100 stream appear to favour the penalty-based approach at this horizon. Seed-level variance is elevated across all methods; notably, `Adam` contains one outlier seed at $\approx$ 14%, pulling the mean below the median. `SGD` (18.7%) and `ONG` (20.1%) barely exceed random chance, indicating near-complete forgetting on this benchmark.

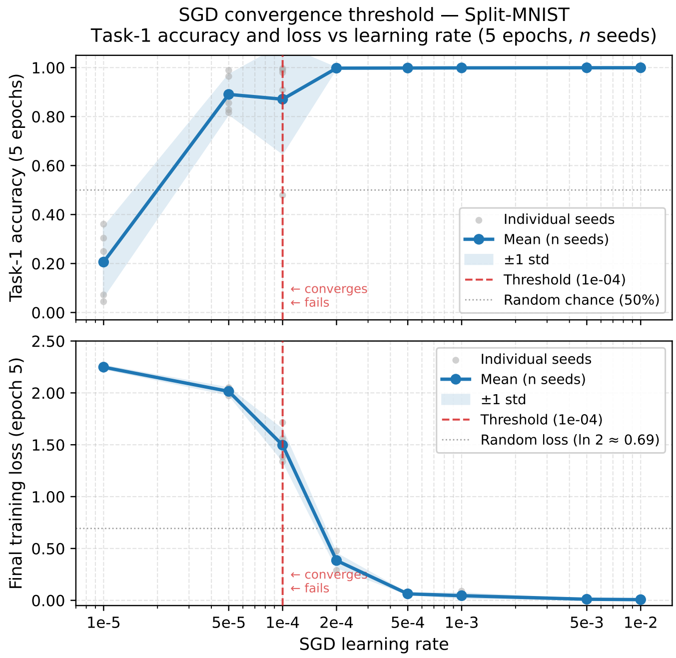

Figure 11: **SGD first-task convergence threshold — Split-MNIST (5 epochs, task-1 only).** Top: task-1 accuracy after five epochs of SGD training as a function of learning rate. Bottom: corresponding final training loss. The red dashed line at $\eta = 10^{-4}$ marks the empirical convergence threshold: below it, accuracy is at or near random chance (50%) and loss remains above 1.5; above it, the network converges reliably to near-perfect task-1 accuracy. The FOPNG paper specifies $\eta = 10^{-5}$ for first-task SGD training, which this sweep shows yields only $\approx 21\%$ accuracy — barely above random chance. The convergence threshold lies between 5×10^{-5} and 10^{-4}. All standalone experiments in this thesis use $\eta = 10^{-3}$, which achieves near-perfect first-task convergence and provides a clean initial projection subspace.

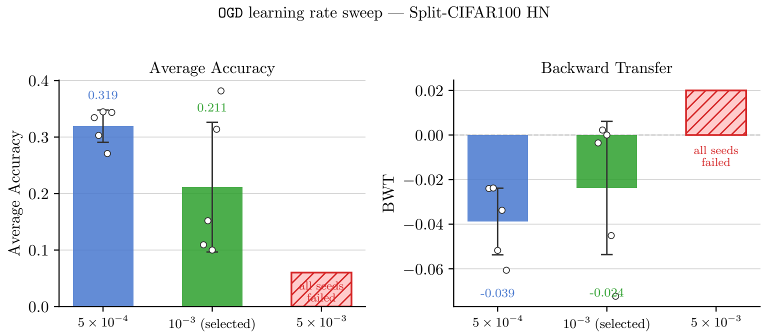

Figure 12: **`OGD` learning rate sweep — Split-CIFAR100 hypernetwork (10 tasks, standard capacity), $n = 5$ seeds per rate.** Left: final average accuracy. Right: final backward transfer. Bars show seed mean; error bars show ± 1 standard deviation; circles show individual seeds. $\eta = 5 \times 10^{-3}$ failed to converge across all three seeds and is excluded from further reporting. $\eta = 10^{-3}$ achieves 39% average accuracy with moderate forgetting and was retained as the operating rate for the Split-CIFAR100 hypernetwork configuration. $\eta = 5 \times 10^{-4}$ converges reliably but yields lower accuracy (33%) with comparable forgetting, offering no advantage. This sweep is the sole instance in which a method-specific learning rate required empirical selection beyond the values specified in Garg et al. (2026) Table 1.

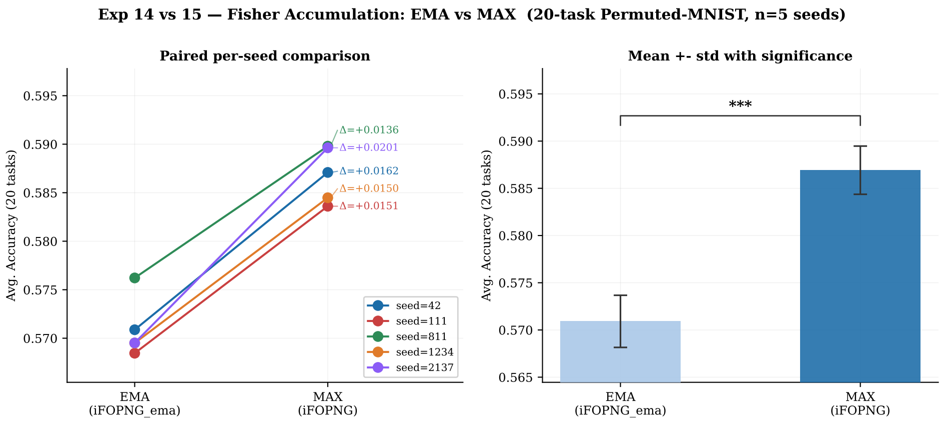

Figure 13: **EMA vs. MAX Fisher accumulation — per-seed statistical comparison, 20-task Permuted-MNIST, $n = 5$ seeds (Sub-RQ4).** Left: paired slope plot. Each line connects the final average accuracy of one seed under EMA (left) and MAX (right); the annotated D values give the per-seed difference. Right: mean $\pm$ std bar chart with the paired t-test significance bracket. Every seed shows a positive slope (MAX $>$ EMA), confirming that the direction of the effect is consistent without exception. The per-seed differences range from $+1.50$ to $+1.65$ pp. Formal tests: paired $t(4) = 14.43$, $p < 0.001$; Wilcoxon signed-rank $W = 15$, $p = 0.031$; Cohen's $d = 7.22$. The null hypothesis that EMA performs at least as well as MAX is rejected by both tests. As discussed in the main text, the effect is statistically unambiguous but practically negligible (1.60 pp mean difference on a benchmark where both strategies lose more than 30 pp over the full sequence).

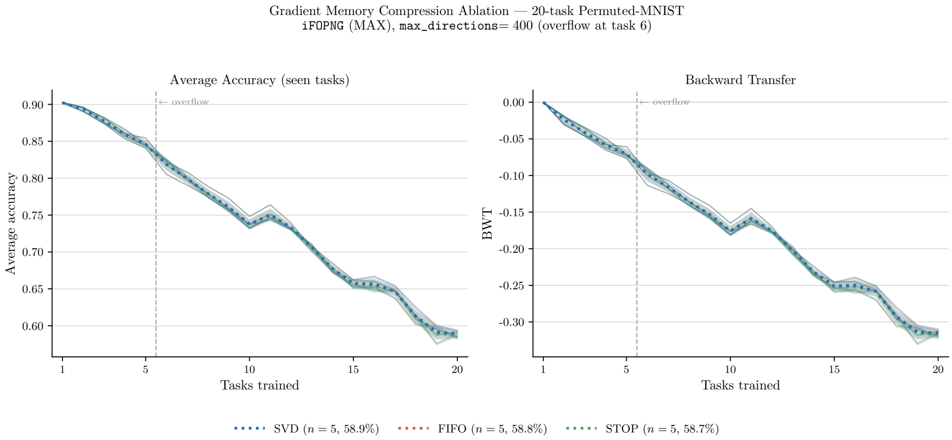

Figure 14: **Gradient memory compression ablation — 20-task Permuted-MNIST, `iFOPNG` with MAX accumulation, `max_directions`**$= 400$, $n = 5$ **seeds each.** Left: average accuracy trajectory over 20 tasks (thick line = seed mean; shaded band = ± 1 standard deviation). Right: backward transfer trajectory from task 2 onward. The vertical dashed line marks the first compression event at task 6, at which point the gradient basis is full and each new task's directions displace existing ones according to strategy. Three strategies are compared: SVD (retain top-K singular vectors), FIFO (evict oldest task directions), and STOP (freeze basis, discard new directions). All three trajectories are visually indistinguishable throughout; final average accuracy spans 58.71%–58.93%, a between-condition range of 0.22 pp. The result confirms that allocation strategy is dominated by budget size: with 400 directions for 20 tasks (≈ 20 per task), the binding constraint is the budget-to-task ratio rather than how the budget is distributed. On Permuted-MNIST, where task distributions are structurally homogeneous, all gradient directions carry equal importance and SVD's importance-weighted selection provides no measurable advantage over the simpler alternatives.

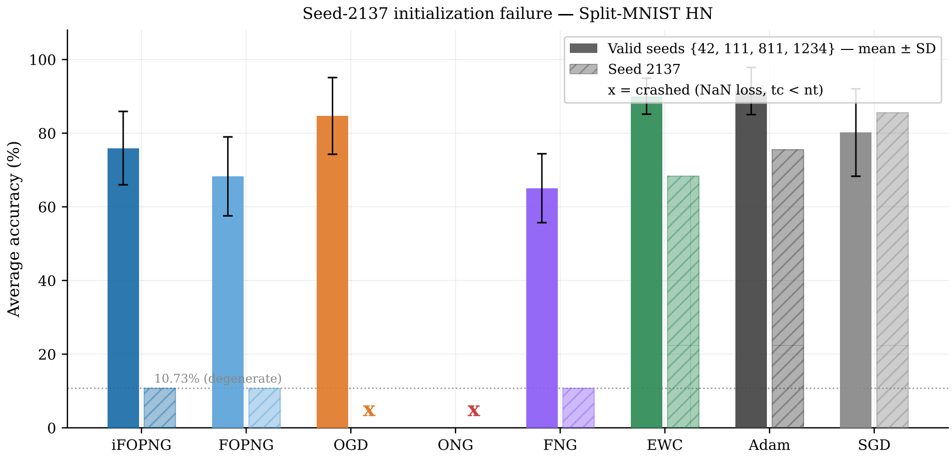

Figure 15: **Seed-2137 initialization failure — Split-MNIST hypernetwork (d_h = 8, 5 tasks, task-incremental single-head).** Each group of bars shows average accuracy for one method; within each group the lighter bar is the mean over the four valid seeds $\{42, 111, 811, 1234\}$ (± 1 SD, shown as error bar) and the darker bar is the value obtained under seed 2137. Methods whose seed-2137 bar is absent (`OGD`, `ONG`) crashed with `NaN` train loss at task 1 and produced no evaluable result. The failure is selective by method class. Unconstrained baselines (`Adam`, `SGD`) and `EWC` show no anomaly under seed 2137, converging normally and posting results within the valid-seed distribution. Every gradient-memory method (`FNG`, `FOPNG`, `iFOPNG`) converges to an identical degenerate accuracy of 10.73% — a value equal to the single-task mean across the five tasks, indicating that task 1 was learned normally (53.7% of final accuracy attributable to task 1 alone) while every subsequent task was never learned. The underlying mechanism is a shared task-1 initialization failure. Under this seed, all adaptive-optimizer runs of the hypernetwork converge to a degenerate task-1 solution at 53.7% accuracy (barely above binary chance on the 5-task single-head protocol) rather than the $\approx 99\%$ reached by SGD on the same seed. Methods that carry nothing forward from task 1 (Adam, EWC) recover by learning tasks 2–5 independently. Methods that bootstrap their gradient memory G and Fisher estimate $\hat{F}$ from the task-1 solution are permanently foreclosed: the projection subspace is seeded from a near-random initialisation, causing all subsequent gradient updates to be projected to near-zero. `OGD` and `ONG` additionally overflow during Fisher estimation on the untrained task-2 embedding, producing `NaN` propagation consistent with the diagnostic in the thesis body. Because the failure is caused by the initialization and affects all projection-based methods identically, it renders any cross-method comparison on this seed invalid; seed 2137 was therefore replaced by seed 314 for the Split-MNIST hypernetwork experiment (see Section 4.6.2).

APPENDIX E: EXTENDED RESULTS

This appendix provides a complete numerical reference for all eight methods across all eight benchmarks evaluated in this thesis. Table 1 reports seed-averaged final average accuracy; formal statistical comparisons are reserved for the targeted `iFOPNG` vs. `FOPNG` analysis in Sub-RQ3 (Section 5.3) and the EMA vs. MAX analysis in Sub-RQ4 (Section 5.4).

Table 1: **Mean average accuracy (%) across all benchmarks and methods.** Values are seed-averaged final average accuracy after all tasks. Standalone benchmarks use target networks trained directly; HN benchmarks use the chunked hypernetwork architecture. †MNIST HN is the suffocated regime ($d_h = 8$). ‡CIFAR-100 MH `EWC` used $\lambda = 10$; $\lambda = 50$ collapsed and is excluded. *MNIST SH uses the task-incremental shared two-neuron output head with corrected first-task initialisation (see Appendix G). **Bold**: best per column. ***Bold italic***: `iFOPNG` (ours). —: insufficient seeds or unavailable.

	Standalone					*Hypernetwork*		
Method	Perm-MNIST	MNIST MH	MNIST SH*	CIFAR-10 MH	CIFAR-100 MH‡	MNIST HN (d_h=8)†	CIFAR-10 HN	CIFAR-100 HN
iFOPNG (ours)	***85.9***	95.9	***80.0***	81.2	42.9	77.2	86.0	37.8
FOPNG	85.7	95.2	78.7	67.1	36.9	65.4	65.4	33.8
EWC	83.9	93.3	76.1	**82.0**	46.6	87.5	83.1	**45.5**
OGD	80.6	**96.5**	66.7	71.3	**49.0**	86.0	79.6	32.3
FNG	68.6	84.6	61.8	69.8	37.5	63.4	81.1	32.5
ONG	61.0	86.8	60.5	68.5	33.8	—	71.4	20.1
SGD	84.3	95.9	63.8	69.0	45.1	78.4	84.1	18.7
Adam	67.1	95.3	62.7	67.9	40.4	**92.6**	**90.4**	40.1
Random chance	50.0	50.0	50.0	50.0	10.0	50.0	50.0	10.0

APPENDIX F: HYPERPARAMETER TABLES

F.1 Standalone Target-Network Experiments

Table 2: Hyperparameters used for each method across all standalone target-network benchmarks. Learning rates for **Adam**, **SGD**, **FNG**, **OGD**, and **FOPNG** are taken from Garg et al. (2026) Table 1; **iFOPNG** uses identical values to **FOPNG**. **ONG** learning rates were selected empirically, as **ONG** is not evaluated in the original work. **EWC** learning rates were selected empirically for Permuted-MNIST and Split-CIFAR10; Split-MNIST and Split-CIFAR100 follow Garg et al. (2026). The damping λ is shared across **ONG**, **FNG**, **FOPNG**, and **iFOPNG**; **OGD** requires no damping. ‡CIFAR-100 EWC: Garg et al. (2026) report $\lambda_{\mathrm{EWC}} = 50$; this value produced collapsed runs in our replication and $\lambda_{\mathrm{EWC}} = 10$ is used instead.

Hyperparameter	**Adam**	**SGD**	**EWC**	**FNG**	**OGD**	**ONG**	**FOPNG**	**iFOPNG**
Permuted-MNIST								
Learning rate (η)	1e-4	5e-3	1e-4	1e-3	5e-3	5e-3	1e-4	1e-4
λ_{EWC}	–	–	10	–	–	–	–	–
λ (damping)	–	–	–	1e-3	–	1e-3	1e-2	1e-2
Grads per task	–	–	–	–	80	80	80	80
Max directions	–	–	–	–	400	400	400	400
Fisher batch size	–	–	full	full	–	full	full	full
Batch size	10	10	10	10	10	10	10	10
# epochs / task	5	5	5	5	5	5	5	5
Split-MNIST (MH and SH)								
Learning rate (η)	1e-5	5e-4	5e-4	1e-3	5e-4	5e-4	1e-5	1e-5
λ_{EWC}	–	–	400	–	–	–	–	–
λ (damping)	–	–	–	1e-3	–	5e-4	5e-4	5e-4
Grads per task	–	–	–	–	80	80	80	80
Max directions	–	–	–	–	400	400	400	400
Fisher batch size	–	–	full	full	–	full	full	full
Batch size	10	10	10	10	10	10	10	10
# epochs / task	5	5	5	5	5	5	5	5
Split-CIFAR10								
Learning rate (η)	1e-3	5e-2	1e-3	1e-2	5e-2	1e-2	1e-3	1e-3
λ_{EWC}	–	–	50	–	–	–	–	–
λ (damping)	–	–	–	1e-3	–	1e-3	1e-3	1e-3
Grads per task	–	–	–	–	80	80	80	80
Max directions	–	–	–	–	400	400	400	400
Fisher batch size	–	–	1024	1024	–	1024	1024	1024
Batch size	10	10	10	10	10	10	10	10
# epochs / task	5	5	5	5	5	5	5	5
Split-CIFAR100								
Learning rate (η)	1e-4	5e-3	1e-2	5e-3	1e-2	1e-2	5e-3	5e-3
λ_{EWC}	–	–	10‡	–	–	–	–	–
λ (damping)	–	–	–	1e-3	–	1e-3	1e-3	1e-3
Grads per task	–	–	–	–	80	80	80	80
Max directions	–	–	–	–	800	800	800	800
Fisher batch size	–	–	1024	1024	–	1024	1024	1024
Batch size	10	10	10	10	10	10	10	10
# epochs / task	10	10	10	10	10	10	10	10

F.2 Hypernetwork Experiments

All HN experiments: `model = HyperNetwork`, `regulizer = True`, $\beta = 0.1$ (`lam = 0.001`), $\eta = 10^{-3}$, $n = 5$ seeds.

Table 3: **Hypernetwork architecture and training parameters.** `c_emb` = chunk embedding dim; `t_emb` = task embedding dim; `max_dir` = maximum gradient memory directions. †The CIFAR-100 HN results are extracted from the 35 completed runs at chunk = 6 000, $d_h = 32$, `normalize=True`; remaining runs are exploratory pilots.

Role / Benchmark	Tasks	ep/task	max_ep	First opt	d_h	chunk	c_emb	t_emb	grads / task	max_dir
CIFAR-10 HN	5	10	50	AdamW	32	6 000	16	16	80	1 000
CIFAR-100 HN†	10	10	50	AdamW	32	6 000	16	16	200	800
MNIST HN (d_h=8, suffocated)	5	15	15	AdamW	8	64	32	32	80	5 000
d_h sweep — $d_h = 4$	5	15	15	AdamW	4	64	32	32	80	5 000
d_h sweep — $d_h = 8$	5	15	15	AdamW	8	64	32	32	80	5 000
d_h sweep — $d_h = 16$	5	15	15	AdamW	16	64	32	32	80	5 000

F.3 Normalization Ablation

All three conditions: Split-CIFAR10 HN, `iFOPNG` only, chunk = 6 000, $d_h = 32$, $\eta = 10^{-3}$. Conditions differ only in which rescaling operations are active: no normalisation (`normalize=False`), gradient-only (QR on G, raw Fisher), and full normalisation (QR on G plus maximum-entry Fisher scaling).

F.4 Sub-RQ4: Fisher Accumulation Operator

Both operators use 20-task Permuted-MNIST, `TargetNetwork`, `normalize = False`, $\eta = 10^{-4}$, $\lambda = 10^{-2}$, 80 grads/task, full Fisher (60 000 samples), batch = 10, 5 epochs/task, $n = 5$ seeds $\{42, 111, 811, 1234, 2137\}$. MAX uses the element-wise maximum; EMA uses $\alpha = 0.5$.

F.5 Compression Ablation

Same setup as F.4 except $\lambda = 10^{-3}$, $\alpha = 0.3$ (internal F_{old} EMA), and `max_dirs` = 400. The three strategies (SVD, FIFO, STOP) are otherwise identical.